

CSE 4621

Machine Learning

Topic: Unsupervised Learning (Clustering)

Source: [DeepLearning.AI](https://deeplearning.ai)

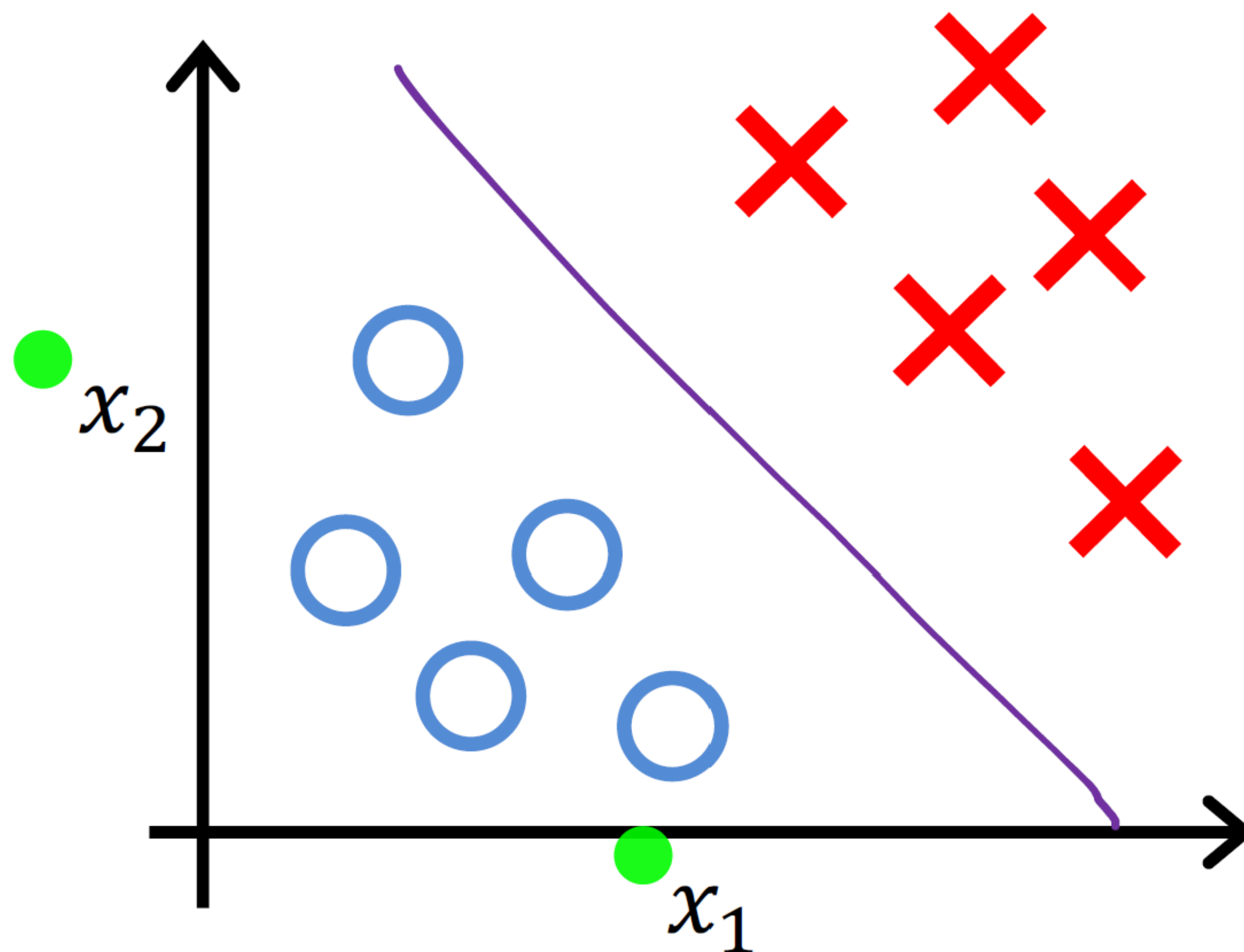
Presented by

Rafsanjany Kushol

Assistant Professor

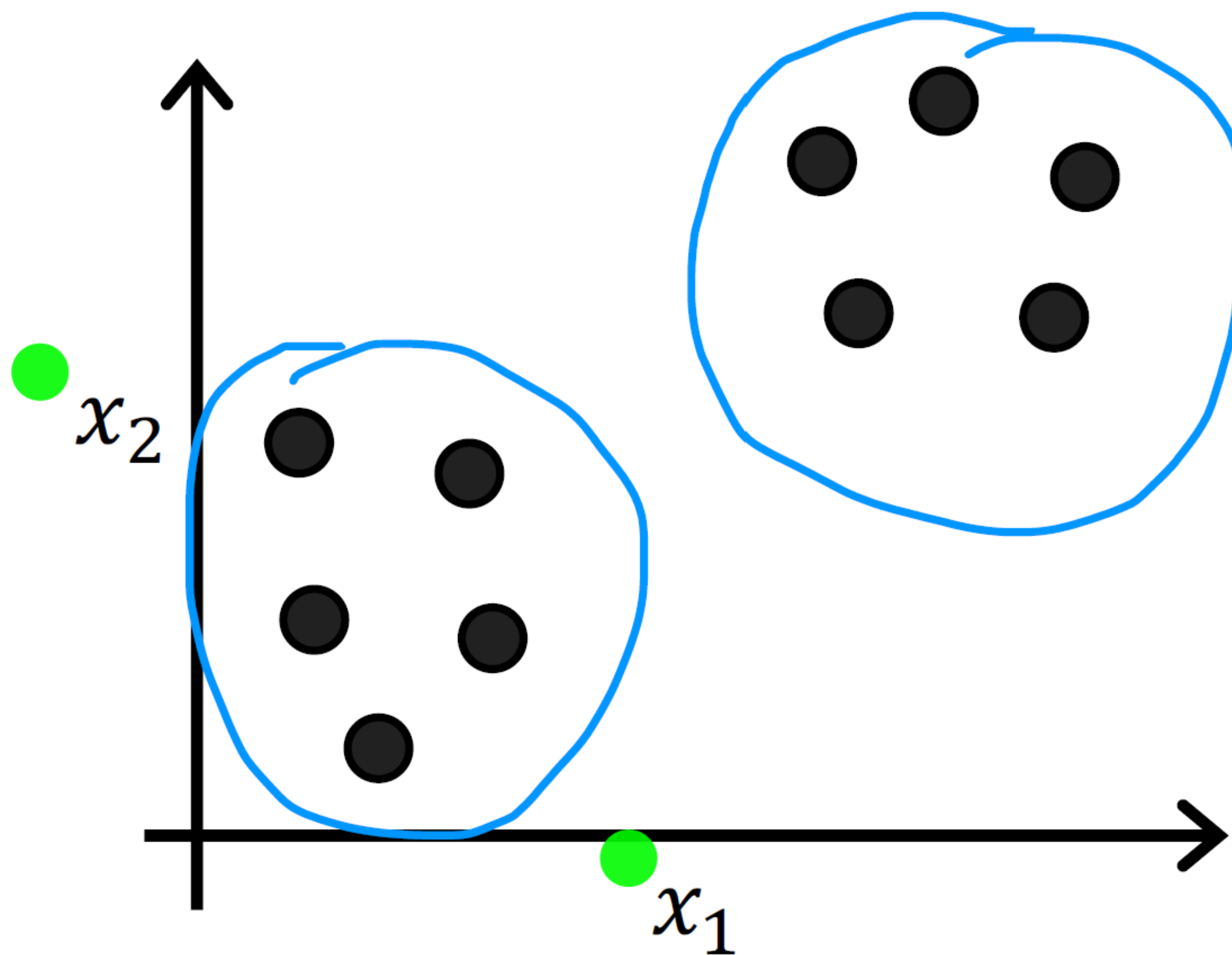


Supervised learning



Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, (x^{(m)}, y^{(m)})\} ?$

Unsupervised learning



Clustering

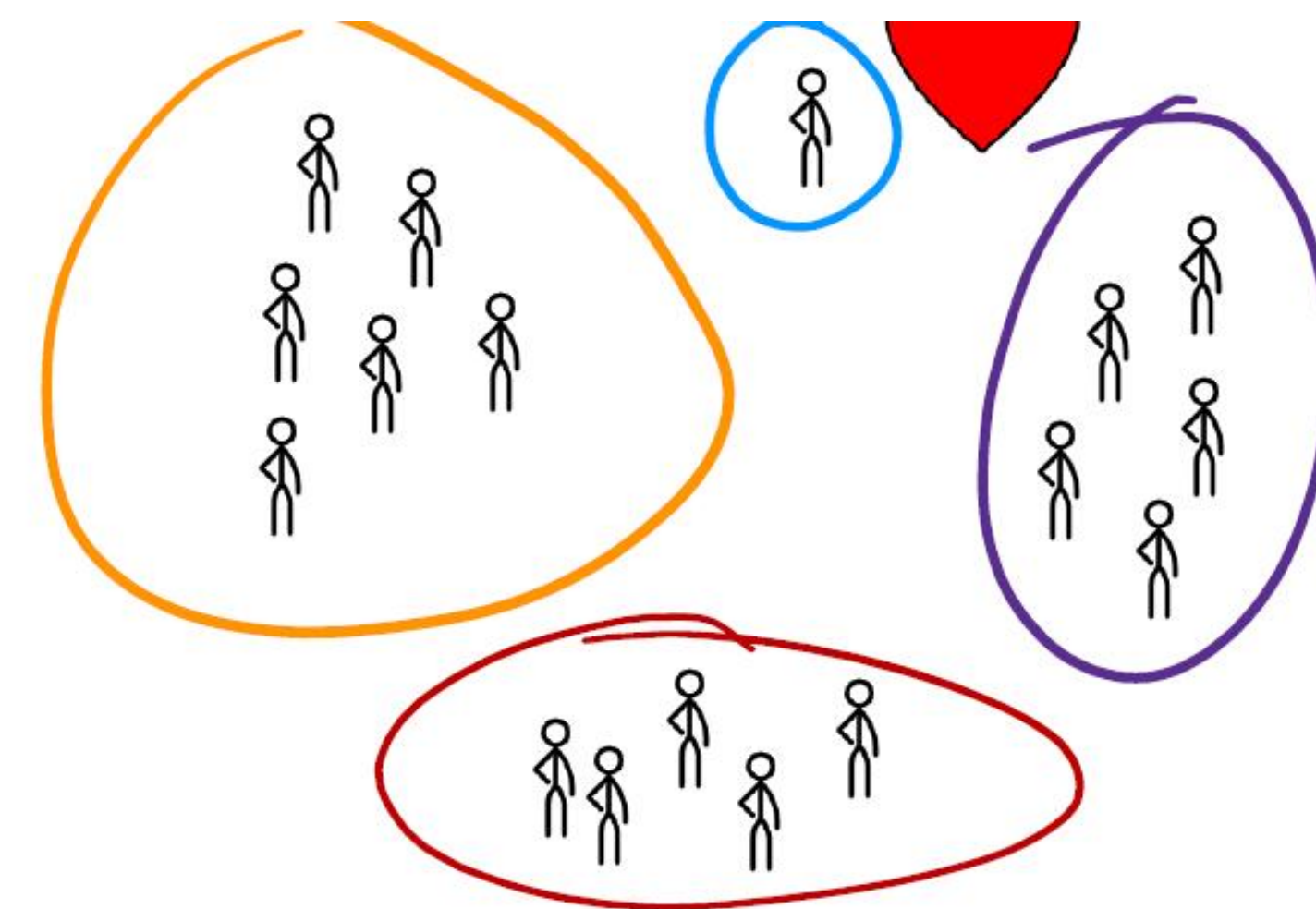
Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(m)}\}$

Applications of clustering

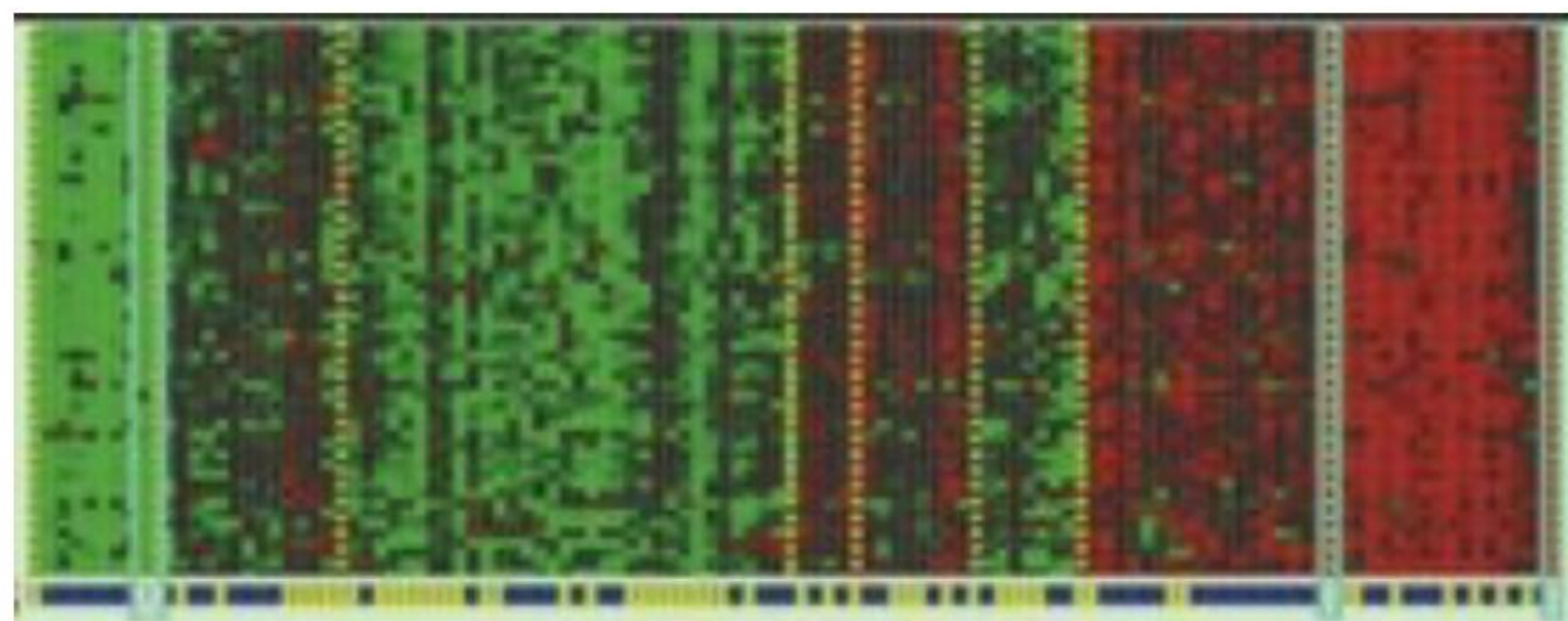


Grouping similar news

- Growing skills
- Develop career
- Stay updated with AI, understand how it affects your field of work



Market segmentation



DNA analysis

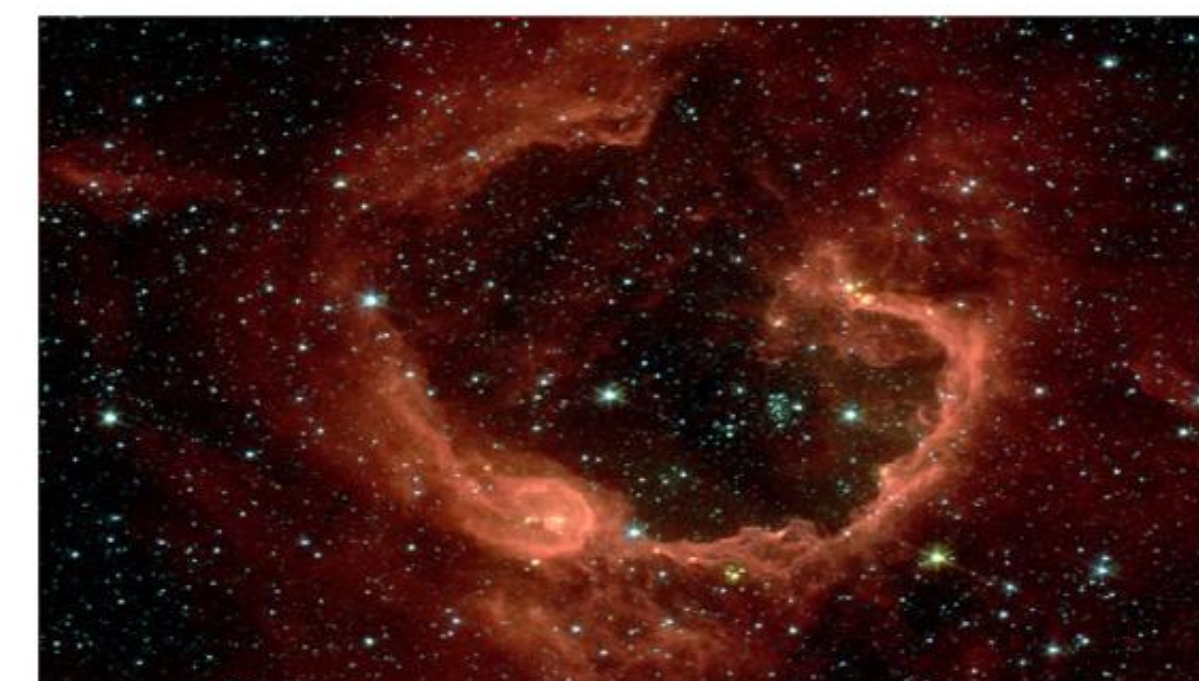


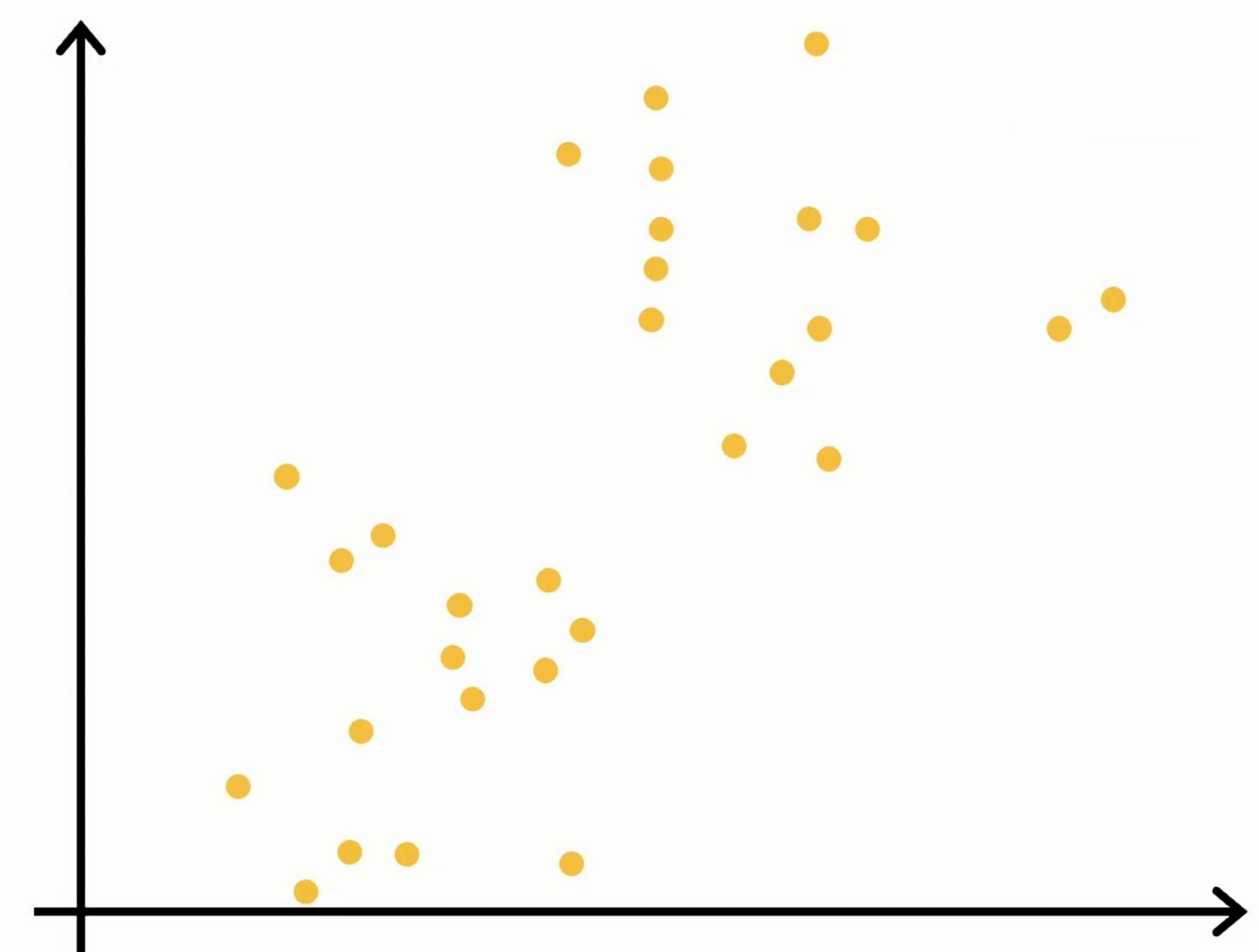
Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis

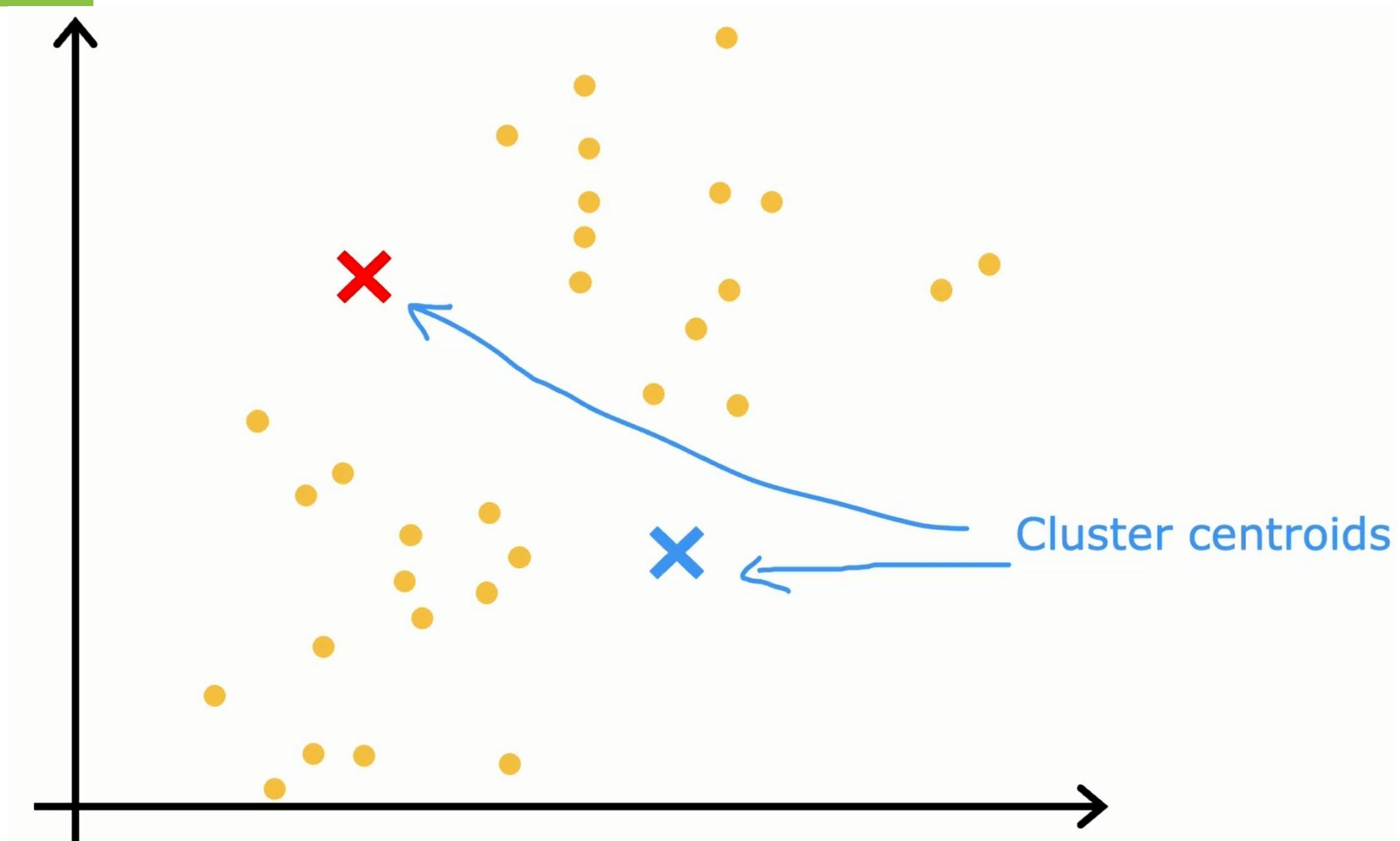
What Is Good Clustering?

- A good clustering method will produce high quality clusters
 - high intra-class similarity: **cohesive** within clusters
 - low inter-class similarity: **distinctive** between clusters
- The quality of a clustering method depends on
 - the similarity measure used by the method
 - its implementation, and
 - Its ability to discover some or all of the hidden patterns

K-means



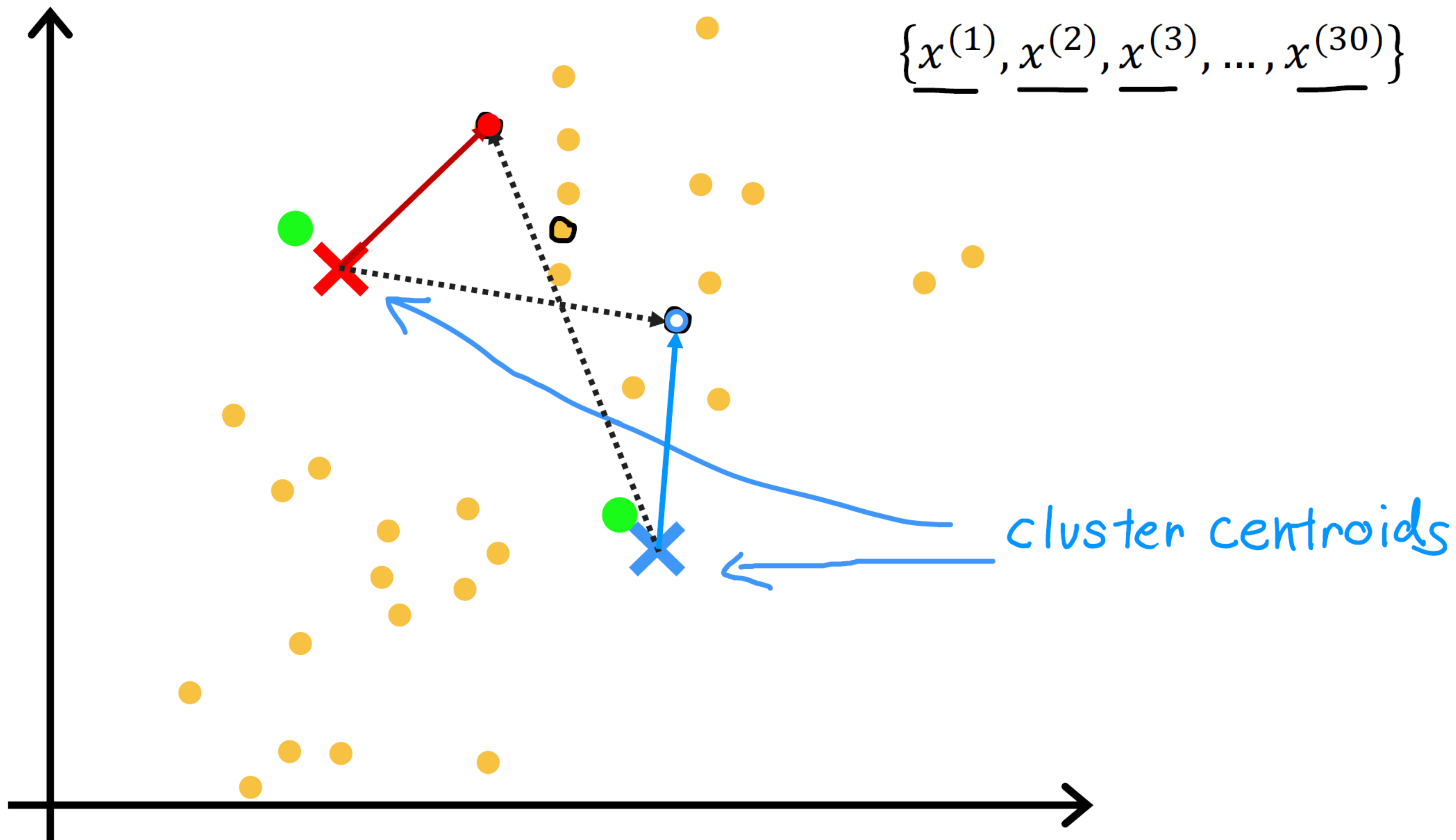
K-means



K-means

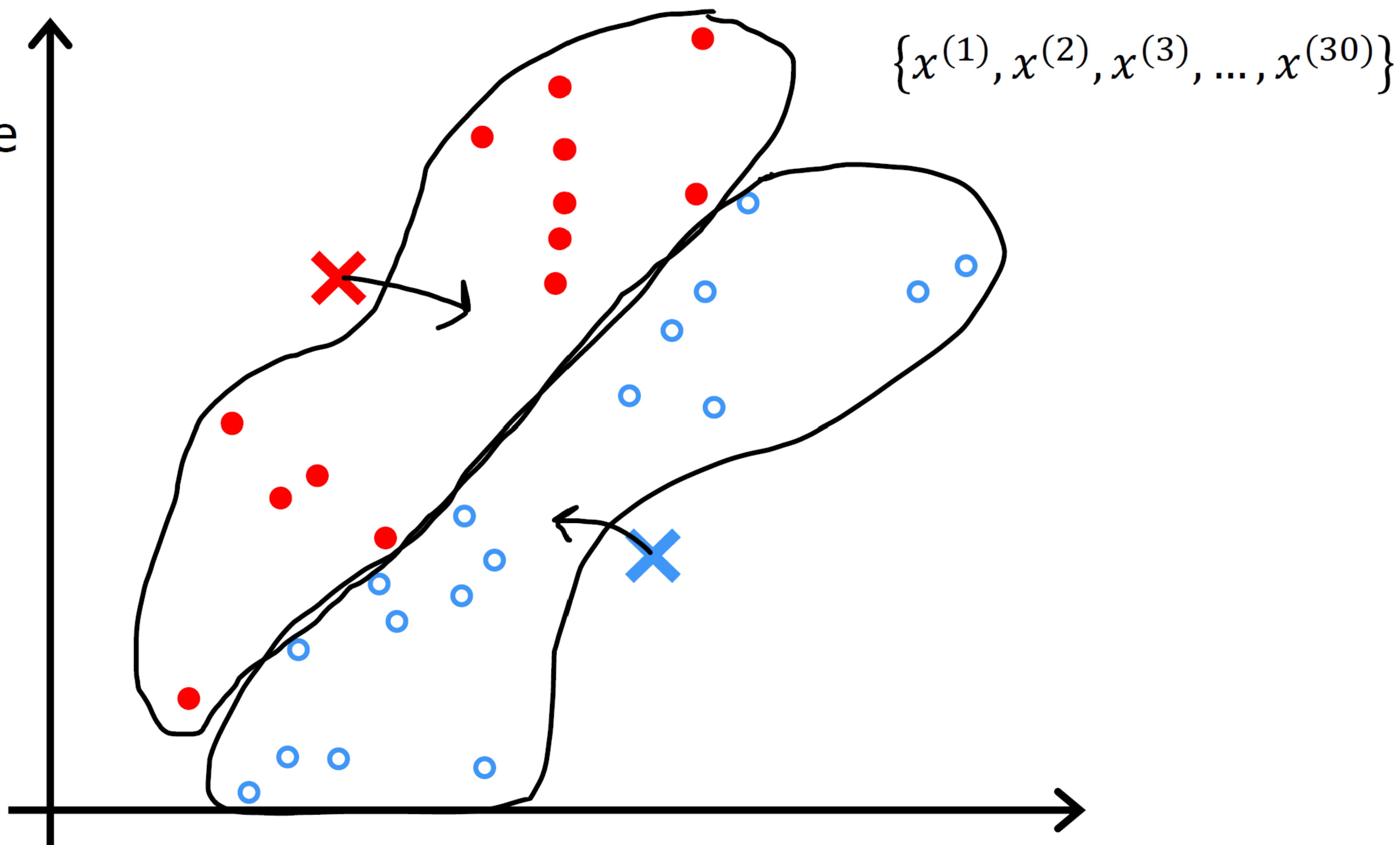
Step 1:

Assign
each point
to its
closest
centroid



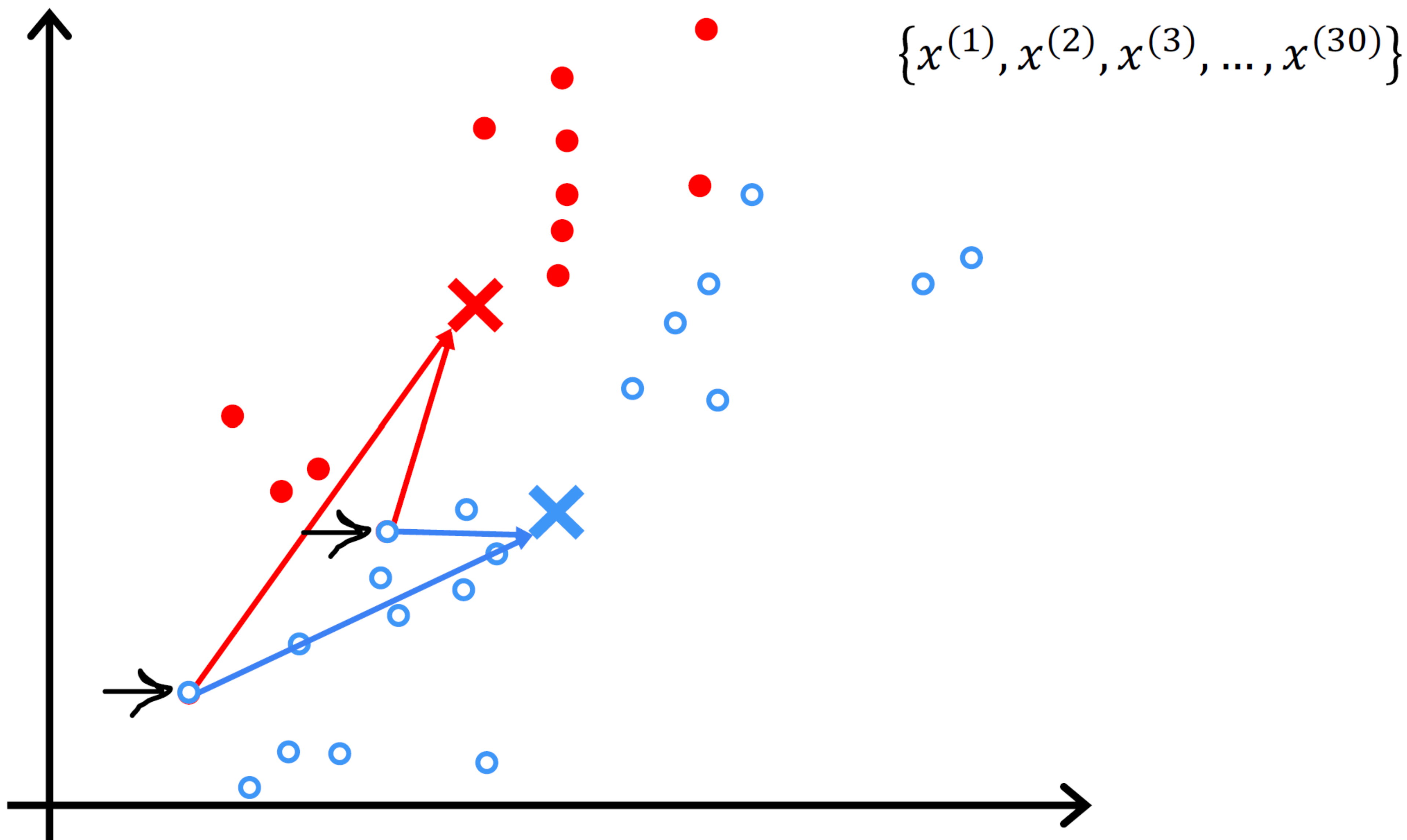
K-means

Step 2:
Recompute
the
centroids



K-means

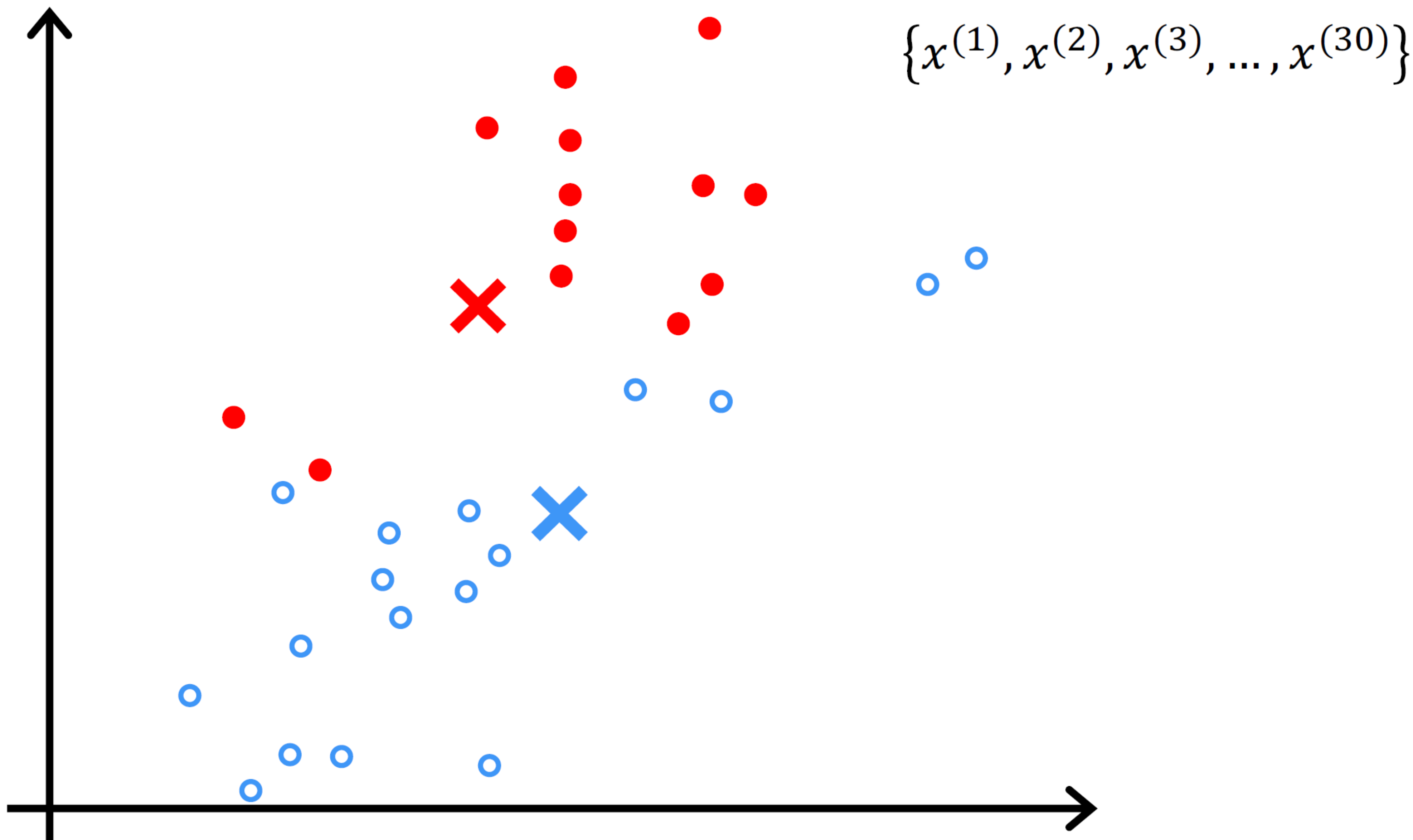
Step 1:
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K-means

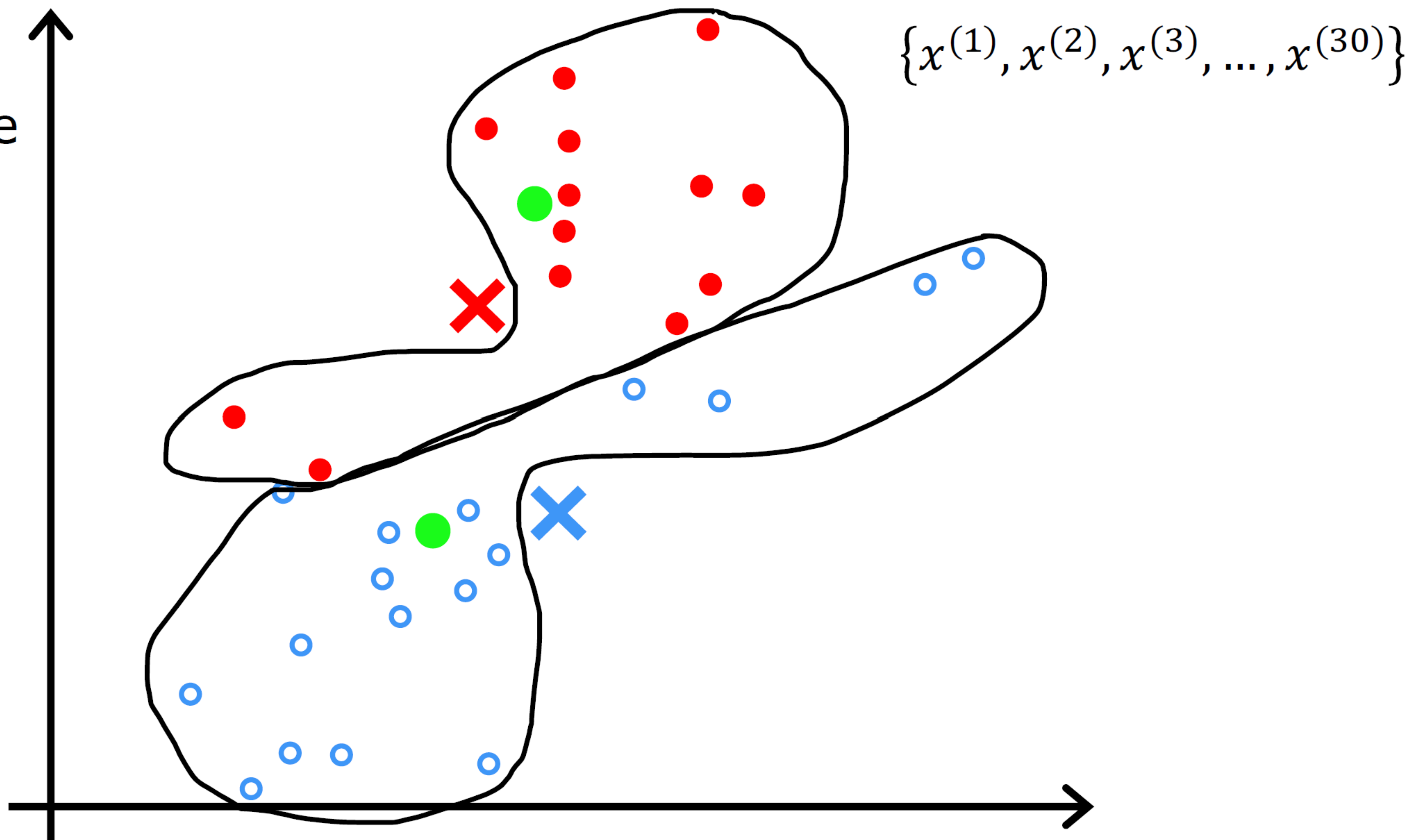
Step 1:

Assign
each point
to its
closest
centroid



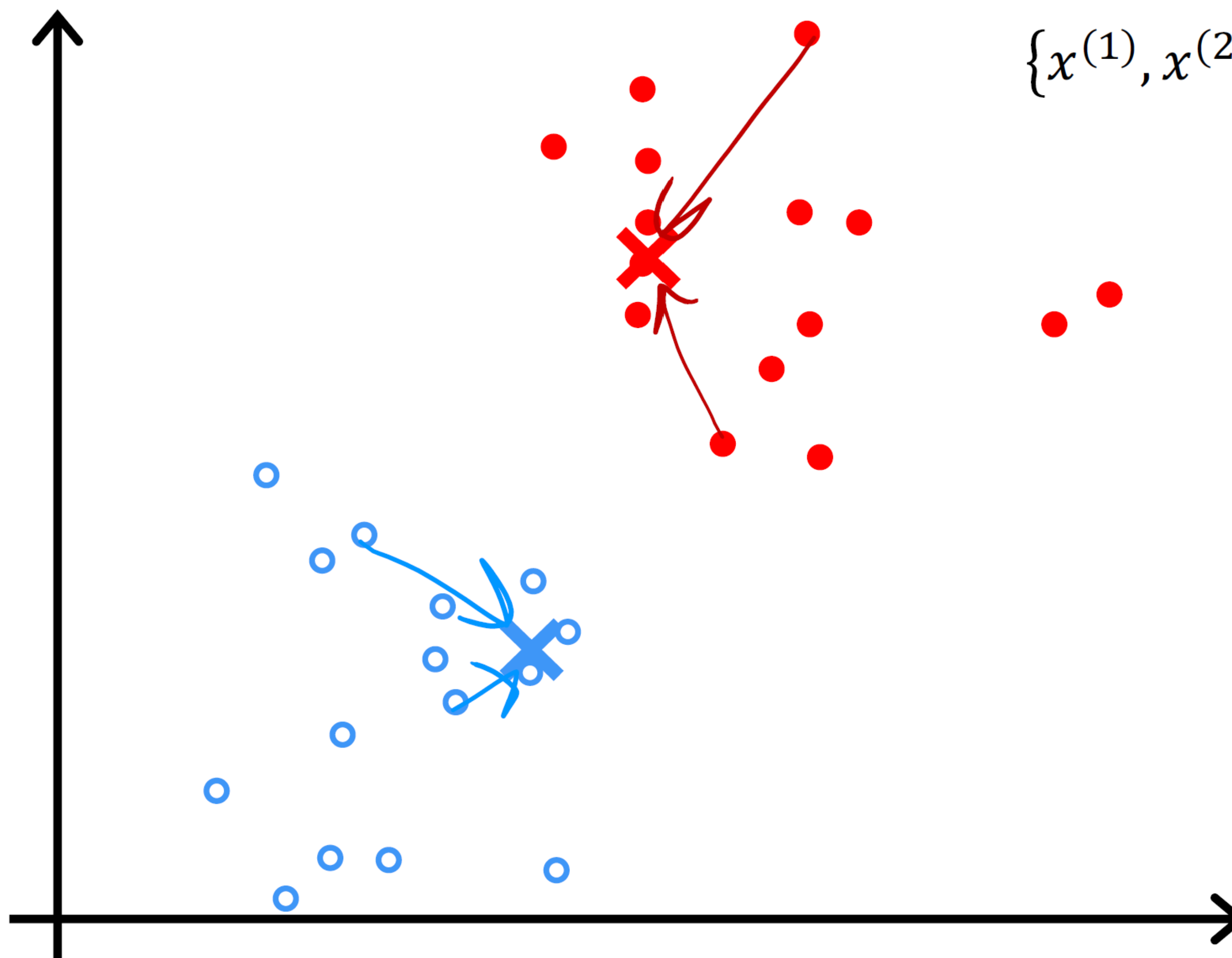
K-means

Step 2:
Recompute
the
centroids



K-means

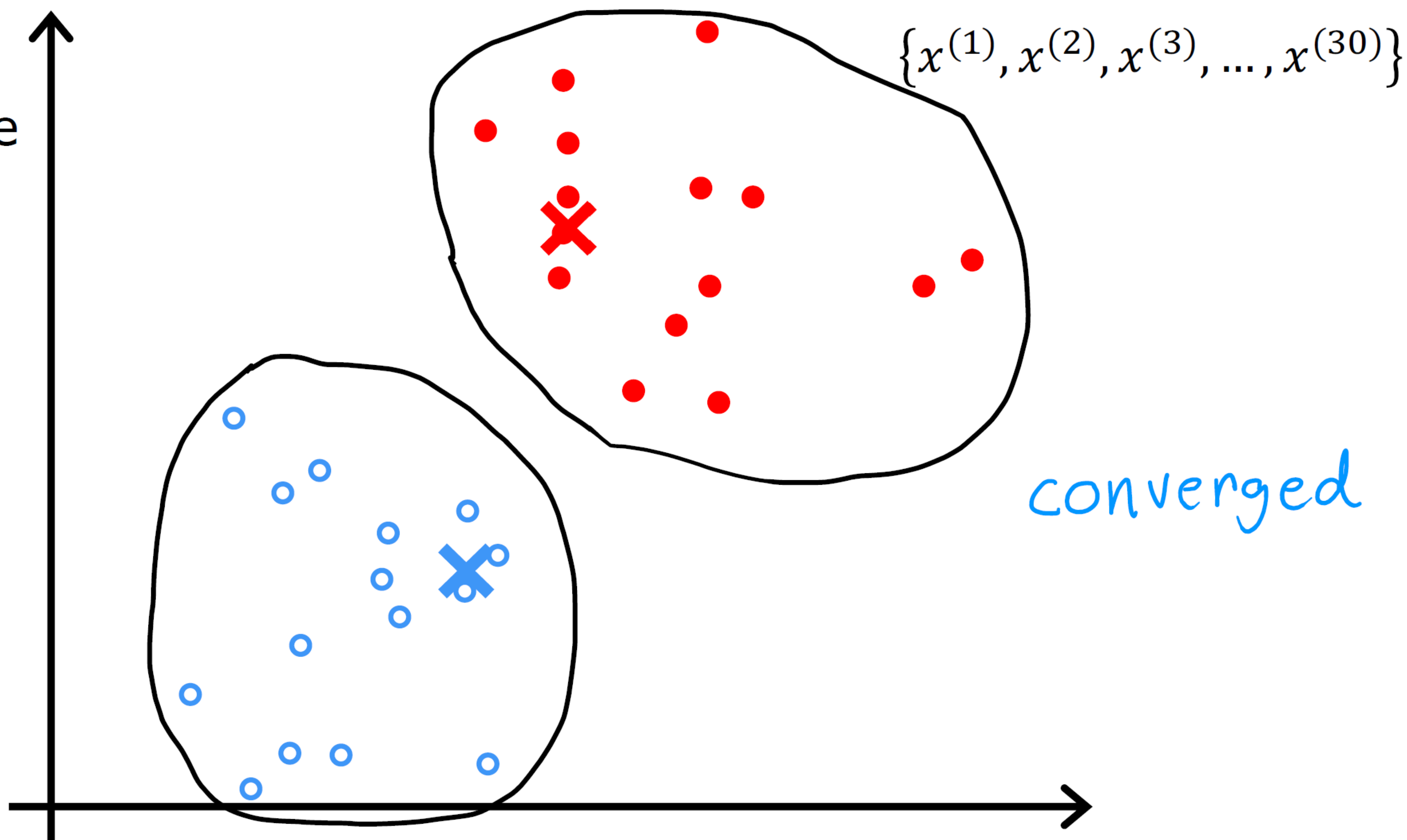
Step 1:
Assign
each point
to its
closest
centroid



$$\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(30)}\}$$

K-means

Step 2:
Recompute
the
centroids

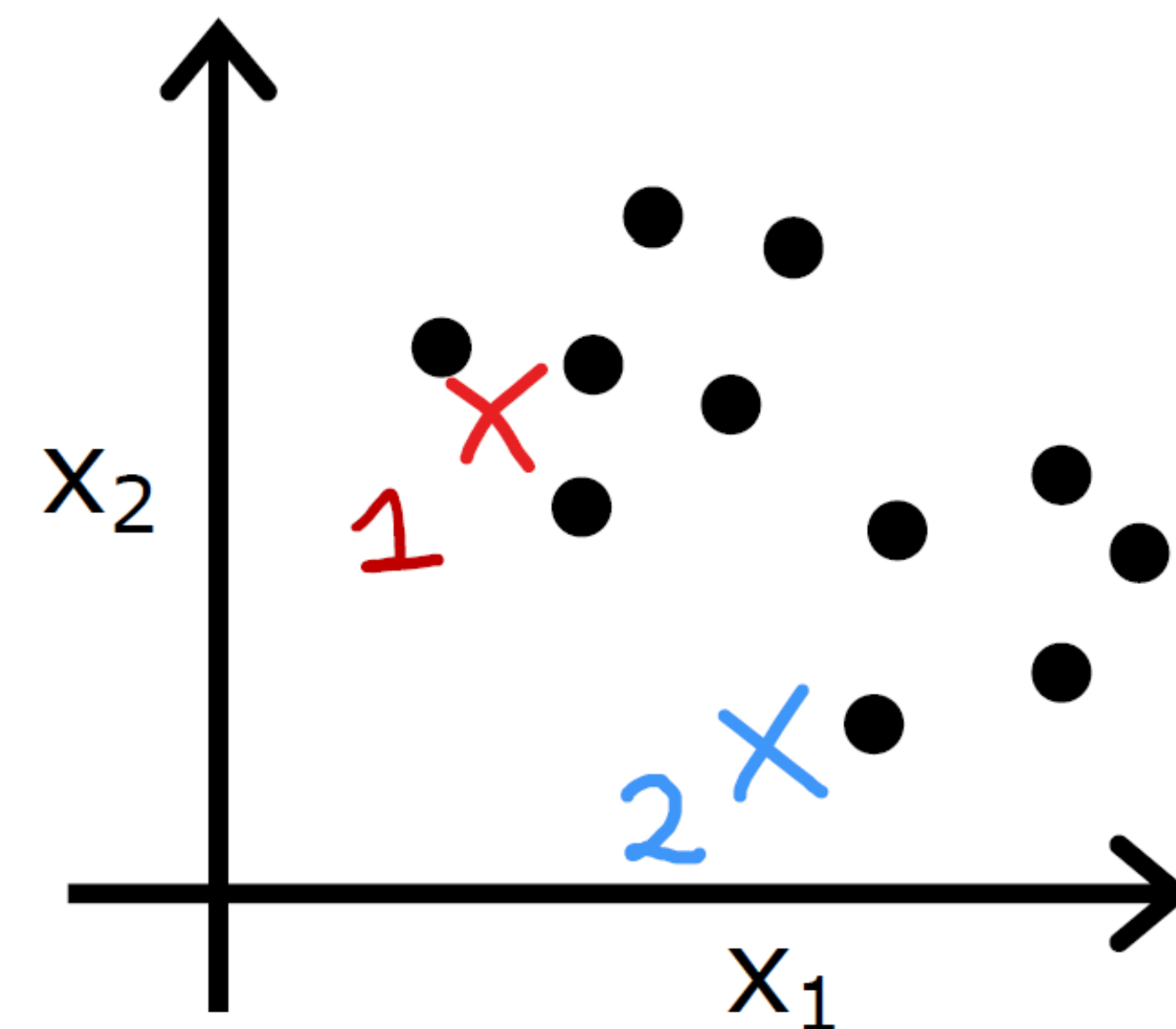


K-means

K-means algorithm

Randomly initialize K cluster centroids

$\mu_1, \mu_2, \dots, \mu_K$
 μ_1, μ_2
 $x^{(1)}, x^{(2)}, \dots, x^{(30)}$
 $n=2$
 $K=2$



K-means

K-means algorithm

$$\mu_1, \mu_2, \dots, \mu_K \quad x^{(1)}, x^{(2)}, \dots, x^{(30)}$$

$$n=2 \quad K=2$$

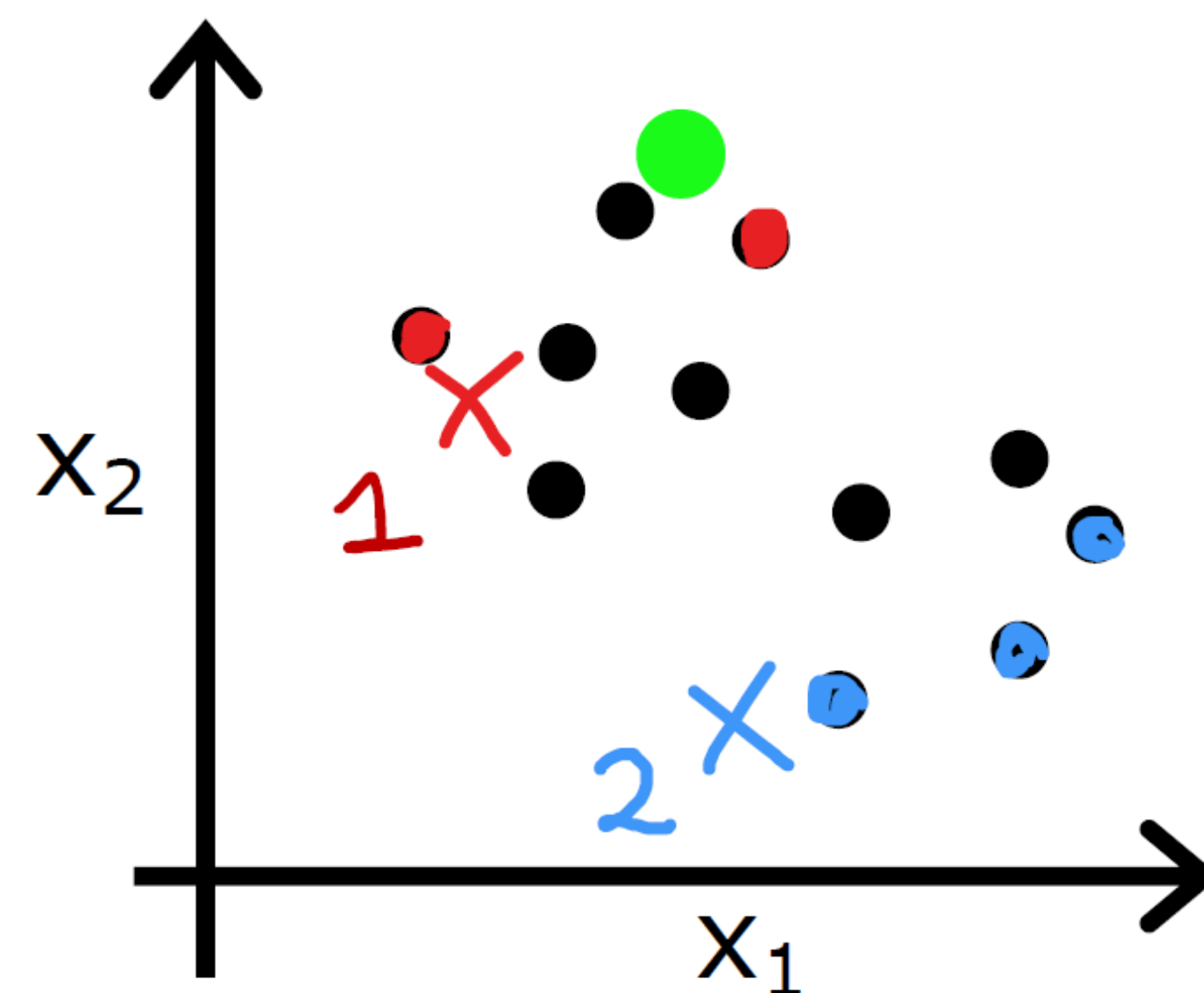
Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K$

Repeat {

Assign points to cluster centroids

for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster centroid closest to $x^{(i)}$ *L2 norm*

$$\min_K \|x^{(i)} - \mu_k\|^2$$


K-means

K-means algorithm

Randomly initialize K cluster centroids

Repeat {

Assign points to cluster centroids

for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster centroid closest to $x^{(i)}$

Move cluster centroids

$$\min_k ||x^{(i)} - \mu_k||^2$$

for $k = 1$ to K

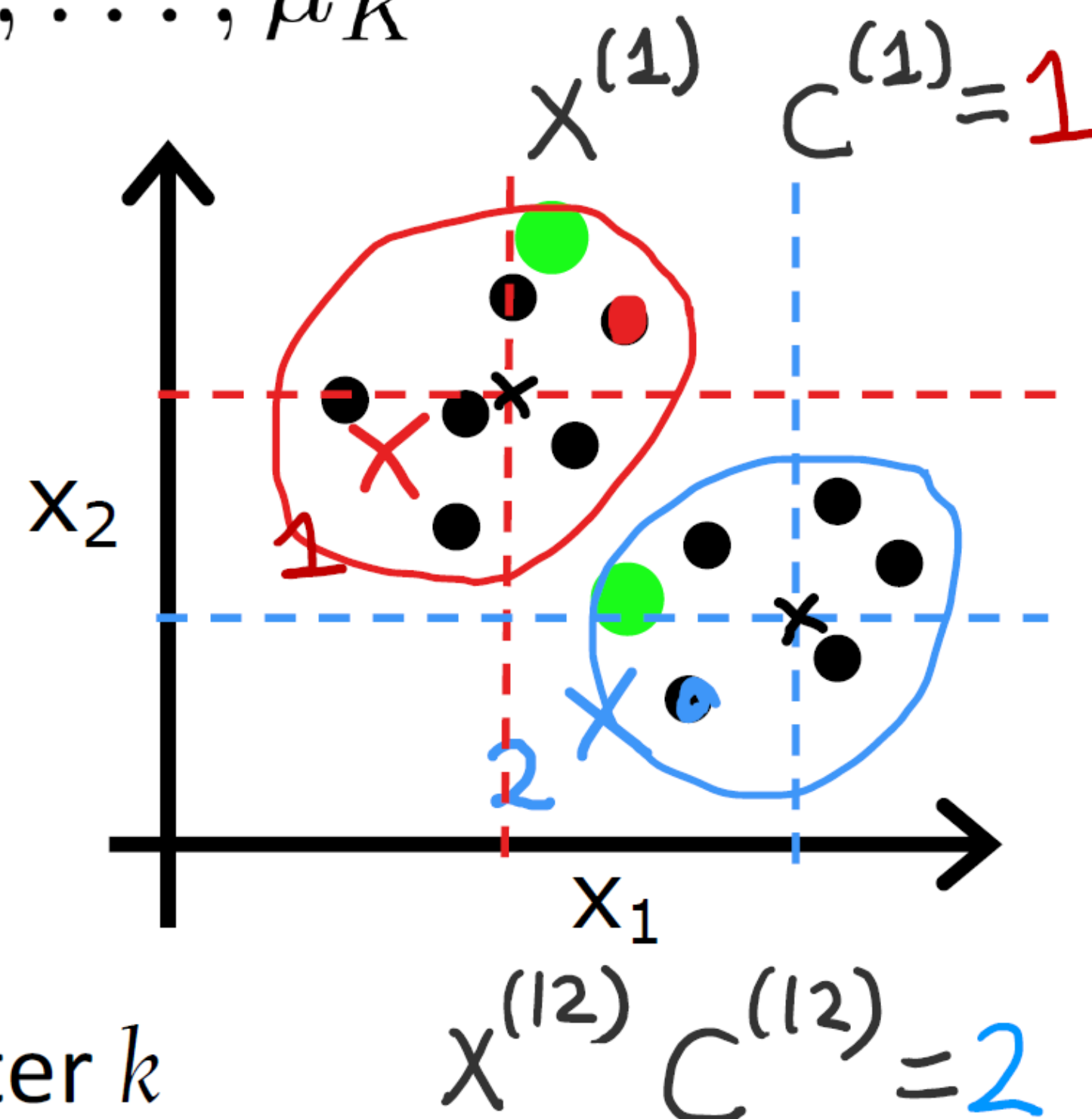
$\mu_k :=$ average (mean) of points assigned to cluster k

}

$$\mu_1 = \frac{1}{4} [x^{(1)} + x^{(5)} + x^{(6)} + x^{(10)}]$$

$$\mu_1, \mu_2, \dots, \mu_K \quad x^{(1)}, x^{(2)}, \dots, x^{(30)} \quad n=2 \quad K=2$$

$$\mu_1, \mu_2, \dots, \mu_K$$



K-means

K-means algorithm

Randomly initialize k cluster centroids, $\mu_1, \mu_1, \dots, \mu_k$

Repeat {

Assign points to cluster centroids

for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster
centroid closest to $x^{(i)}$

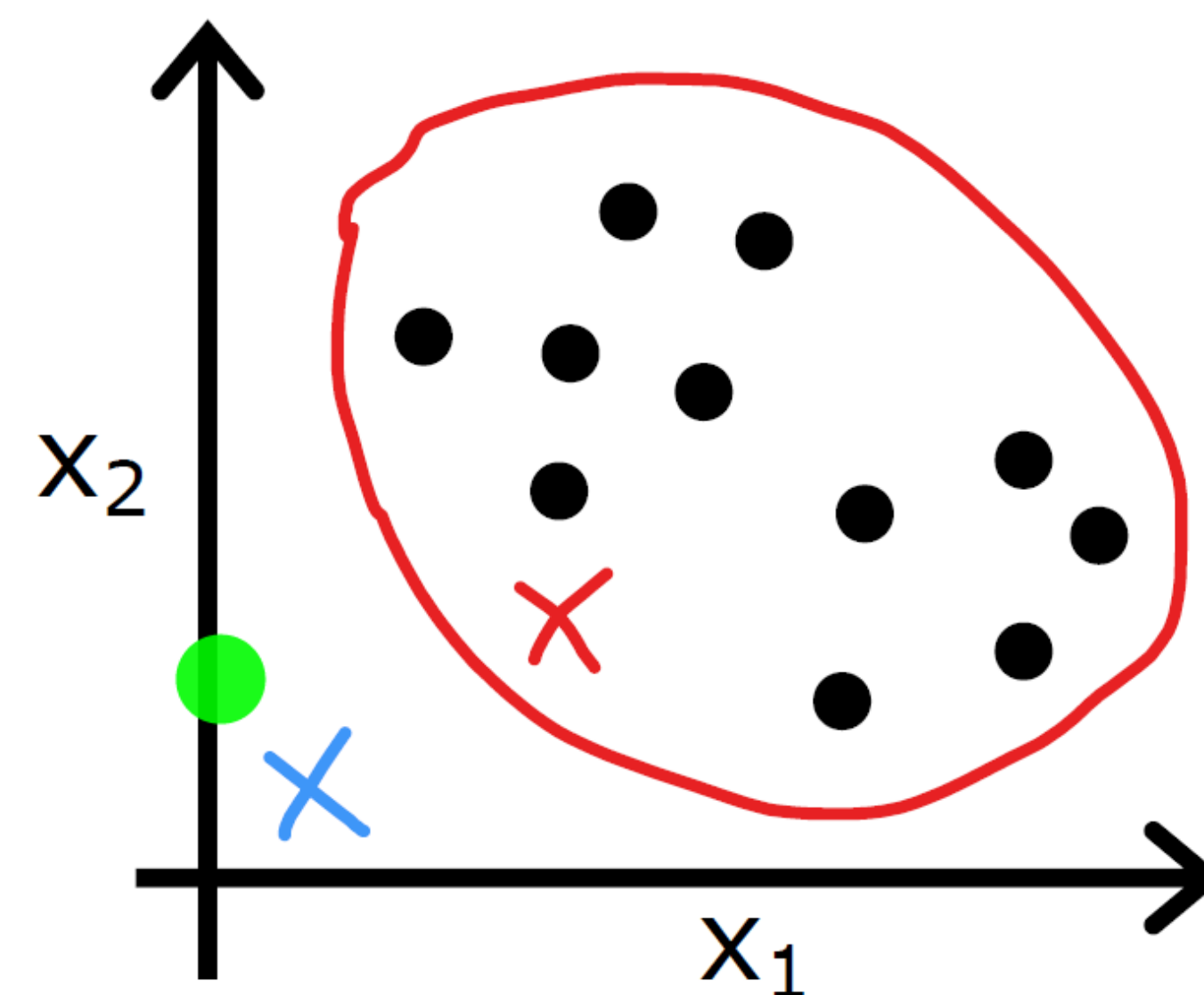
Move cluster centroids

for $k = 1$ to K

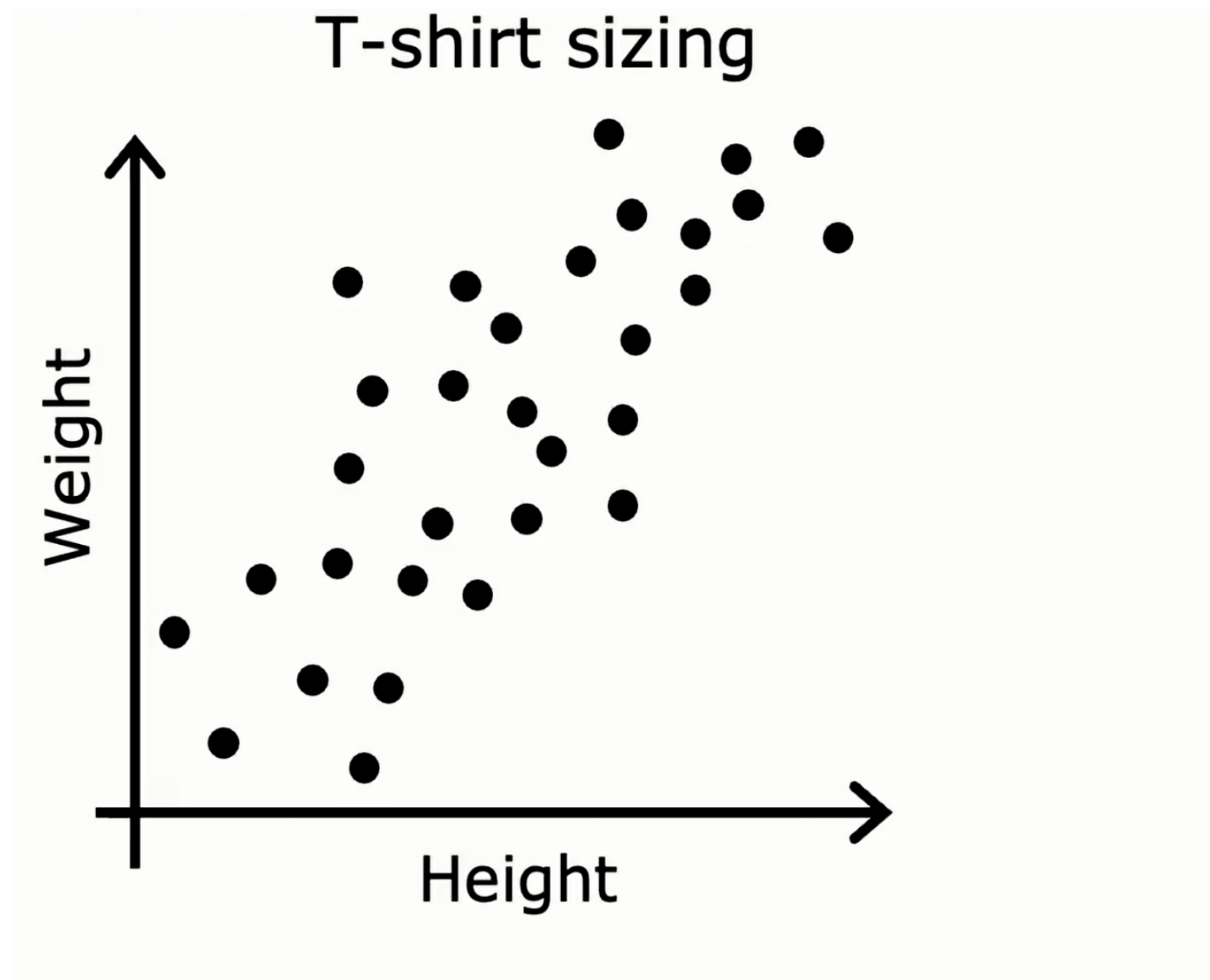
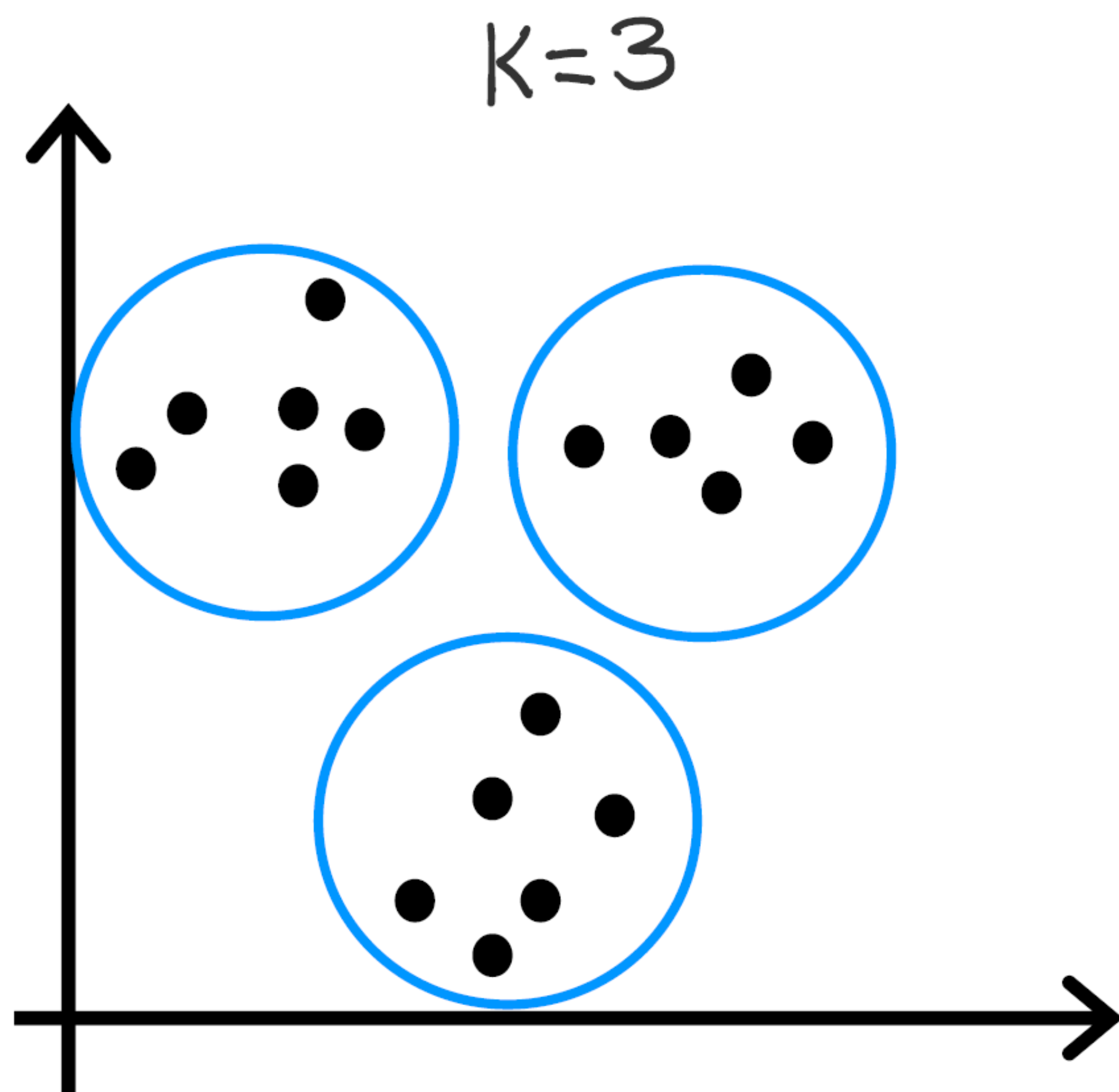
$\mu_k :=$ average (mean) of points assigned to cluster k

}

μ_1, μ_2 $x^{(1)}, x^{(2)}, \dots, x^{(30)}$
 $n=2$ ~~$K=2$~~
 $K = K - 1$ ✓



K-means for clusters that are not well separated



K-means optimization objective

$c^{(i)}$ = index of cluster $(1, 2, \dots, K)$ to which example $x^{(i)}$ is currently assigned

μ_k = cluster centroid k

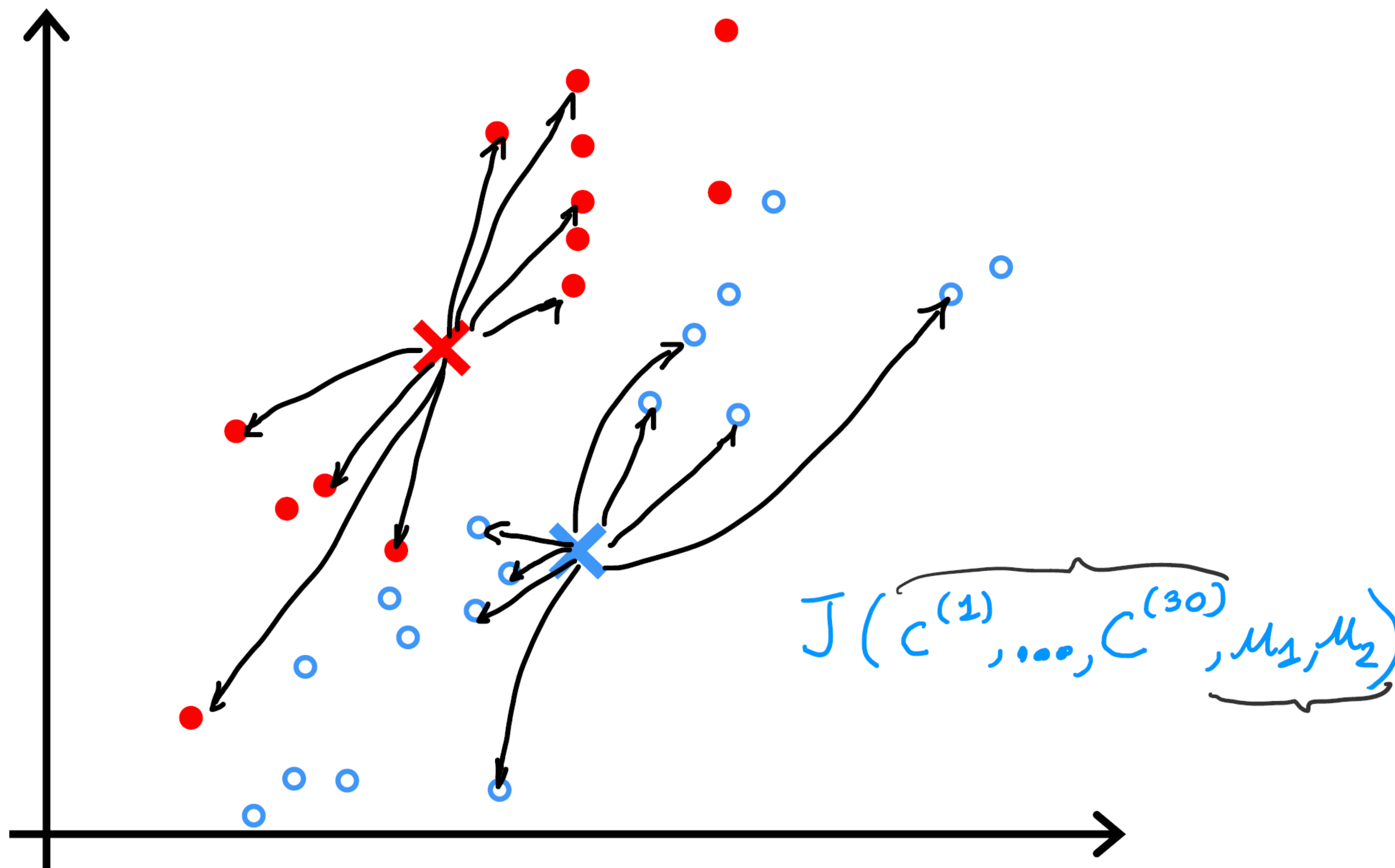
● $\mu_{c^{(i)}}$ = cluster centroid of cluster to which example $x^{(i)}$ has been assigned

Cost function

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

● $\min_{c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K} J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$ distortion

K-means



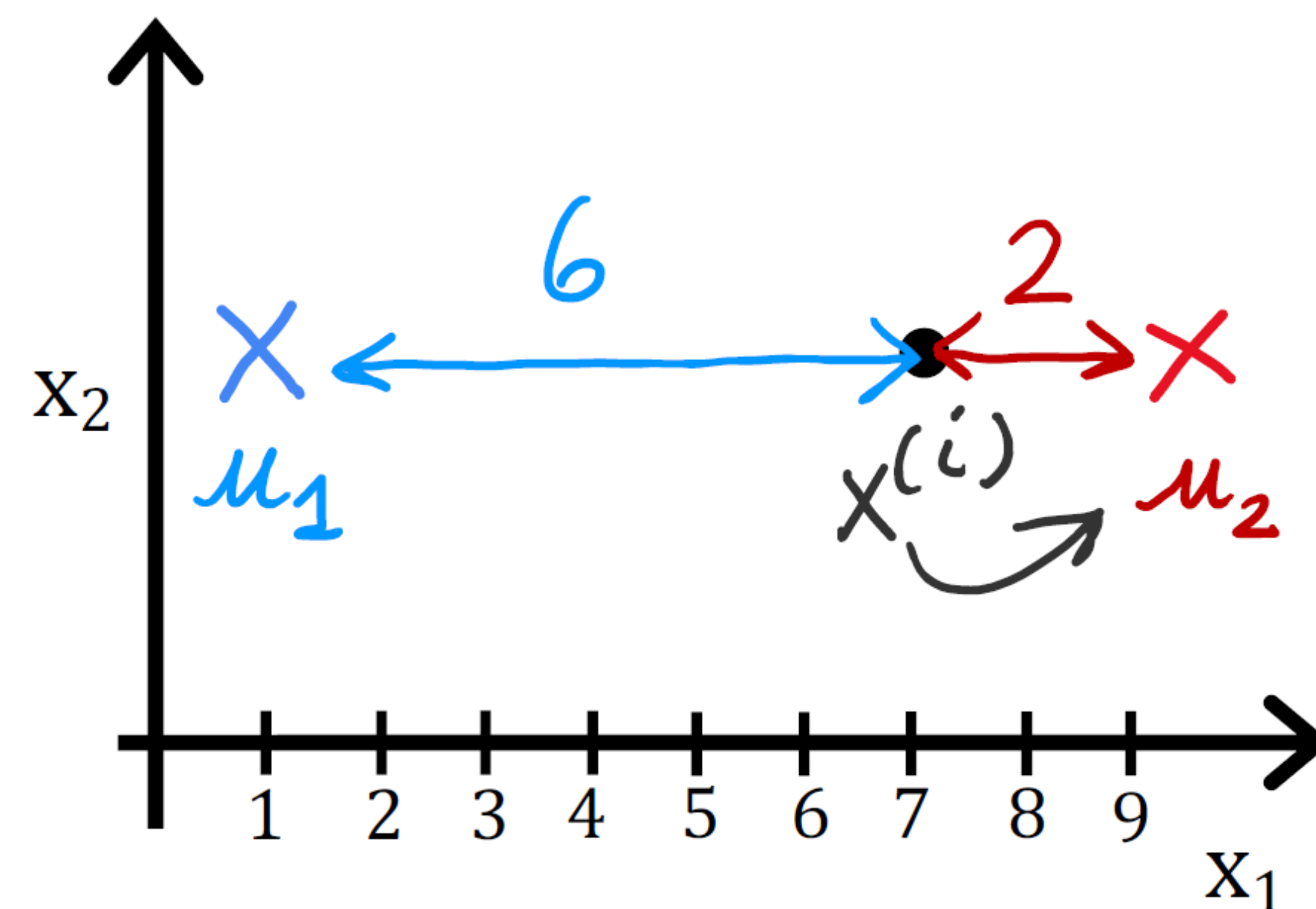
K-means optimization objective

Cost function for K-means

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

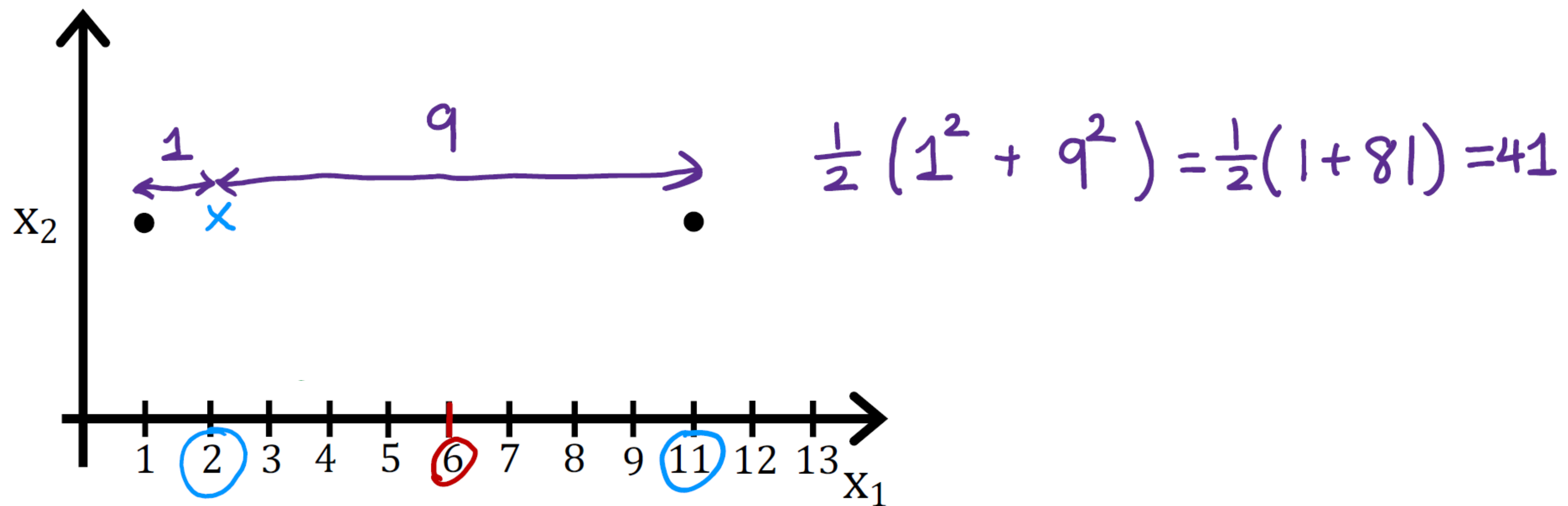
Repeat {

- # *Assign points to cluster centroids*
for $i = 1$ to m :
 - $c^{(i)}$ = index of cluster centroid closest to $x^{(i)}$
- # *Move cluster centroids*
for $i = 1$ to K :
 - μ_k = average of points in cluster



K-means optimization objective

Moving the centroid



K-means algorithm

- Step 0: Randomly initialize K cluster centroids $\mu_1, \mu_1, \dots, \mu_k$

Repeat {

Step 1: Assign points to cluster centroids

Step 2: Move cluster centroids

}

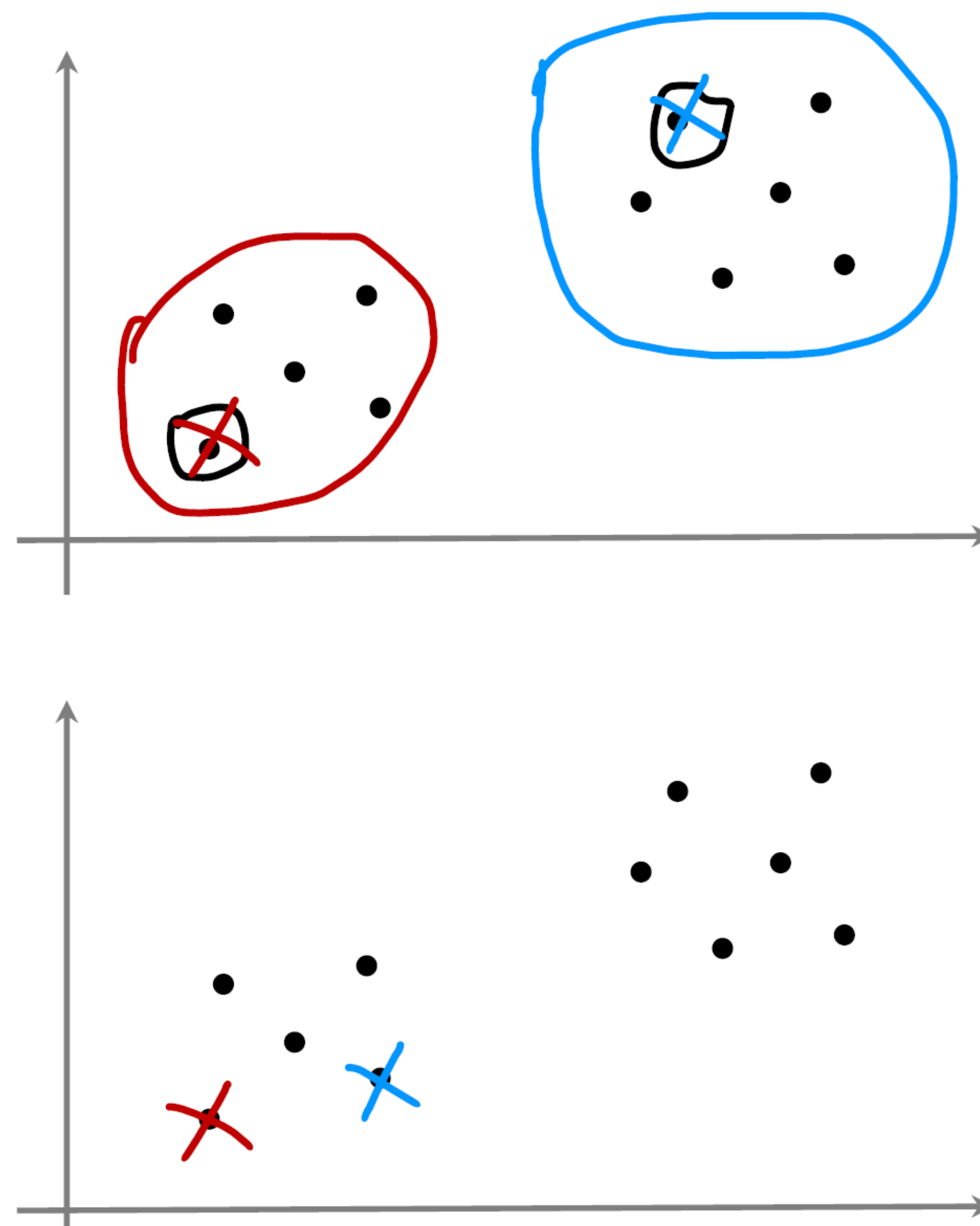
K-means algorithm

Random initialization

Choose K < m

Randomly pick K training examples.

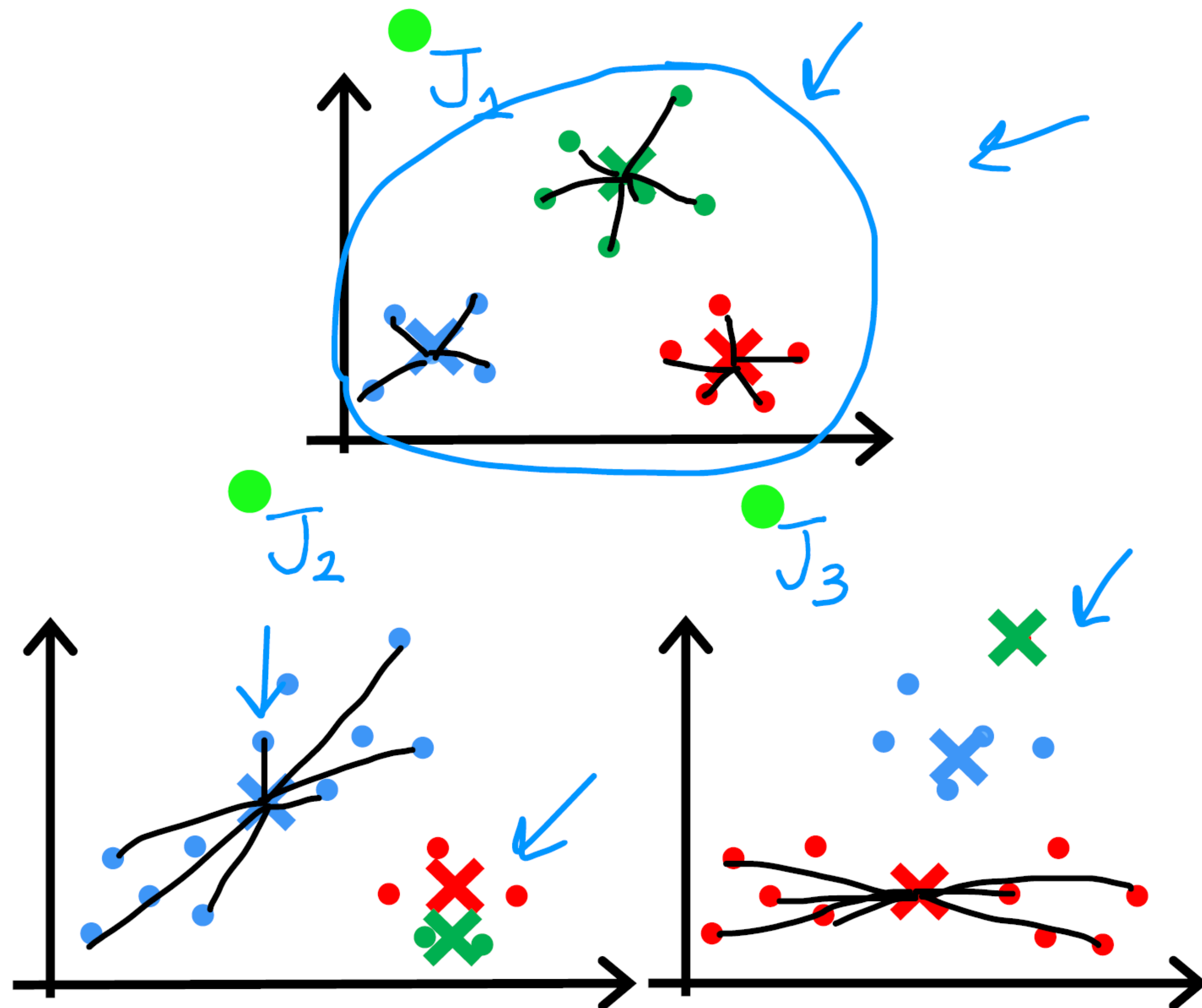
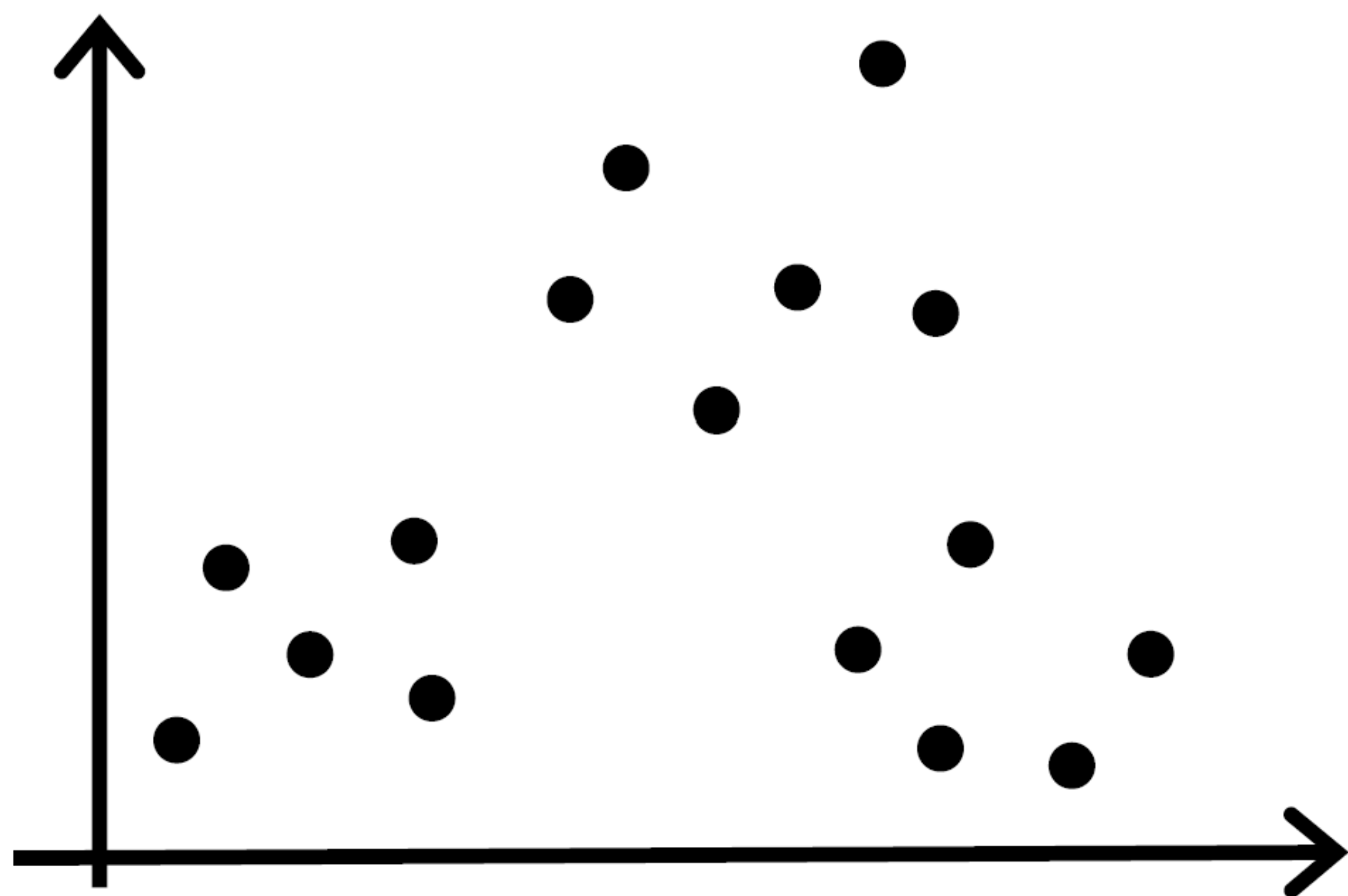
Set $\mu_1, \mu_1, \dots, \mu_k$ equal to these K examples.



Random initialization

$K=3$

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$$



K-means algorithm

Random initialization

For $i = 1$ to 100 { ← 50-1000

better than running K-means once

Randomly initialize K-means. pick K random examples

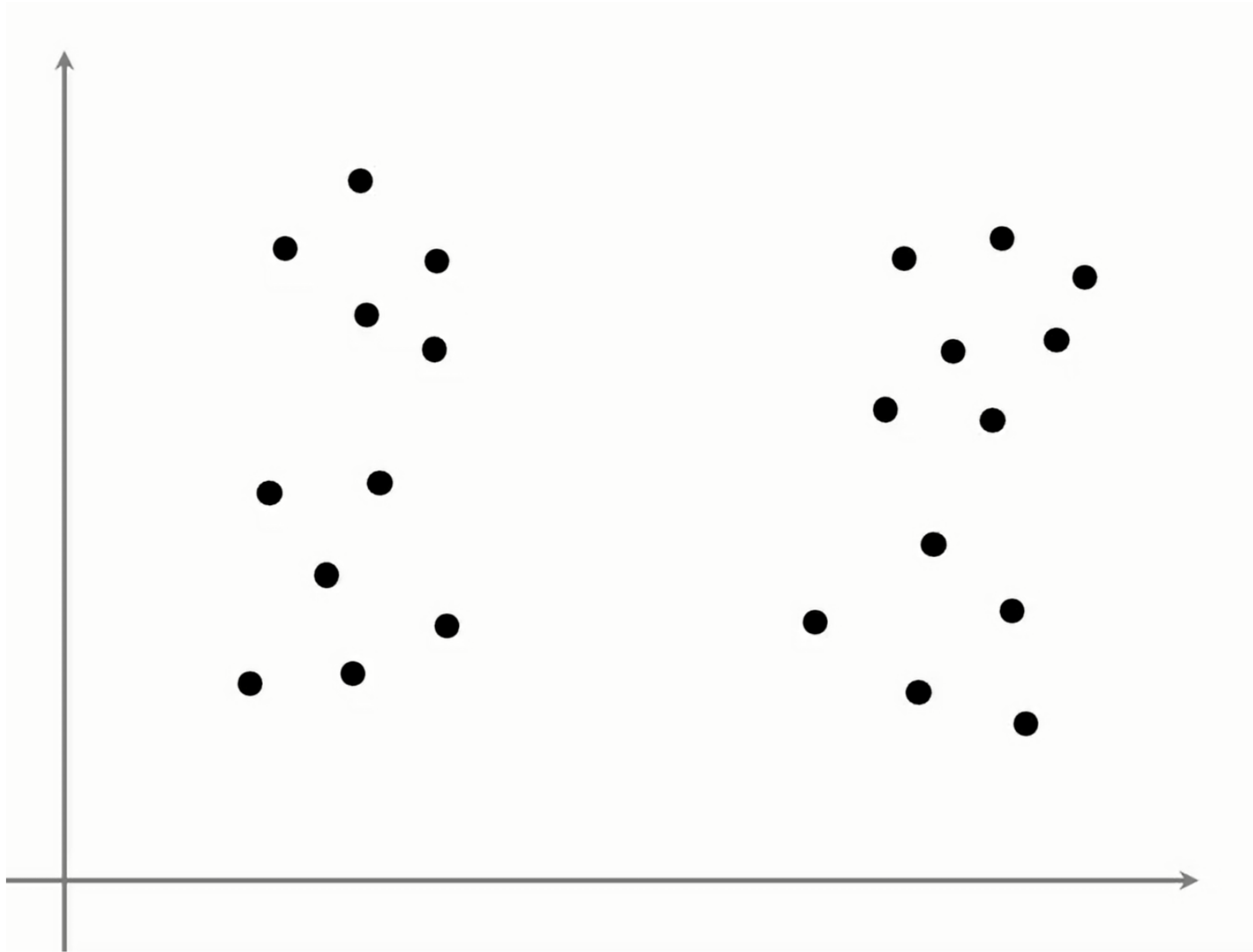
Run K-means. Get $c^{(1)}, \dots, c^{(m)}, \mu_1, \mu_1, \dots, \mu_k$ Set as cluster centroids

Compute $J(c^{(1)}, \dots, c^{(m)}, \mu_1, \mu_1, \dots, \mu_k)$ distortion (cost function)

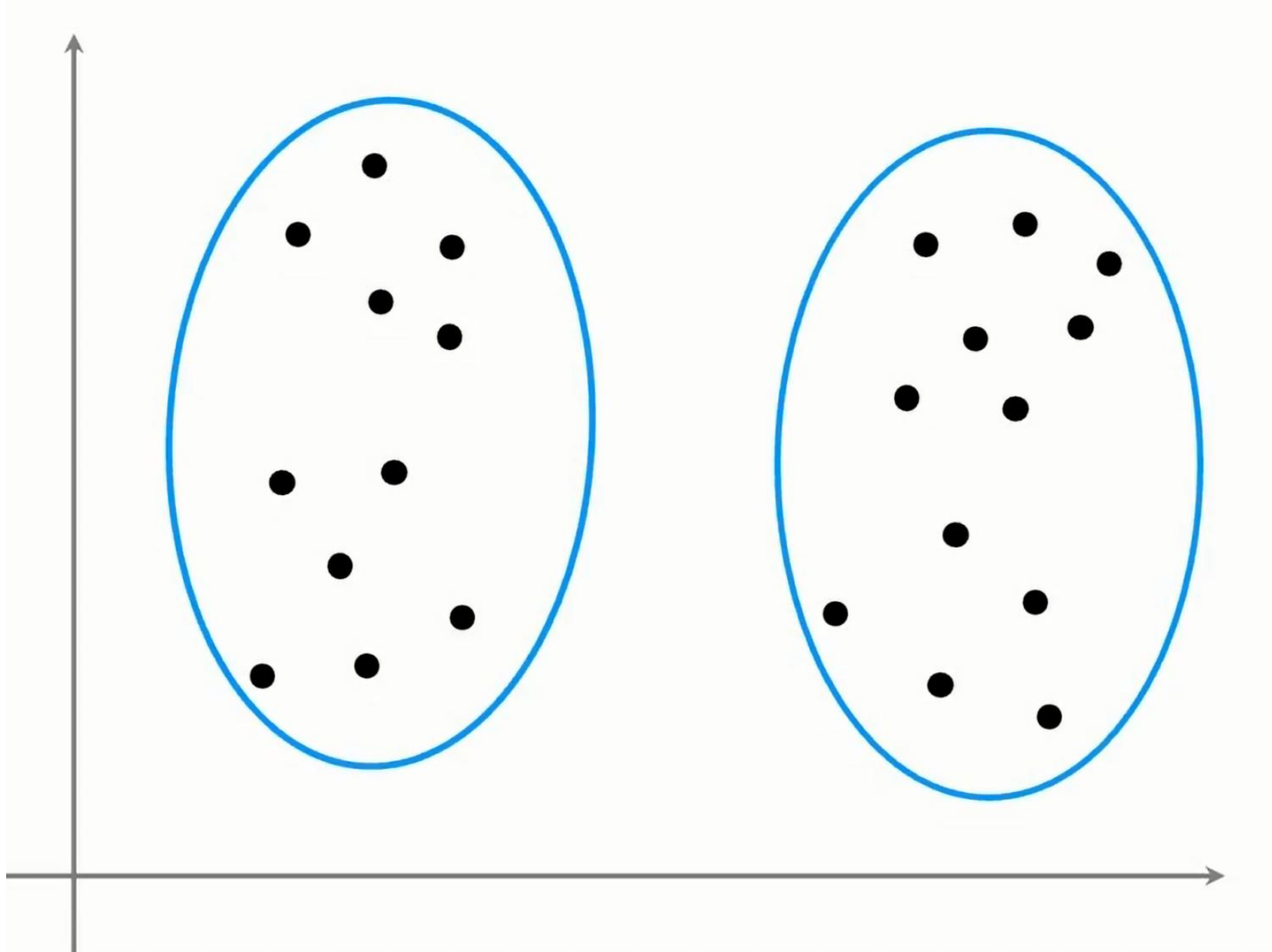
}

Pick set of clusters that gave lowest cost J

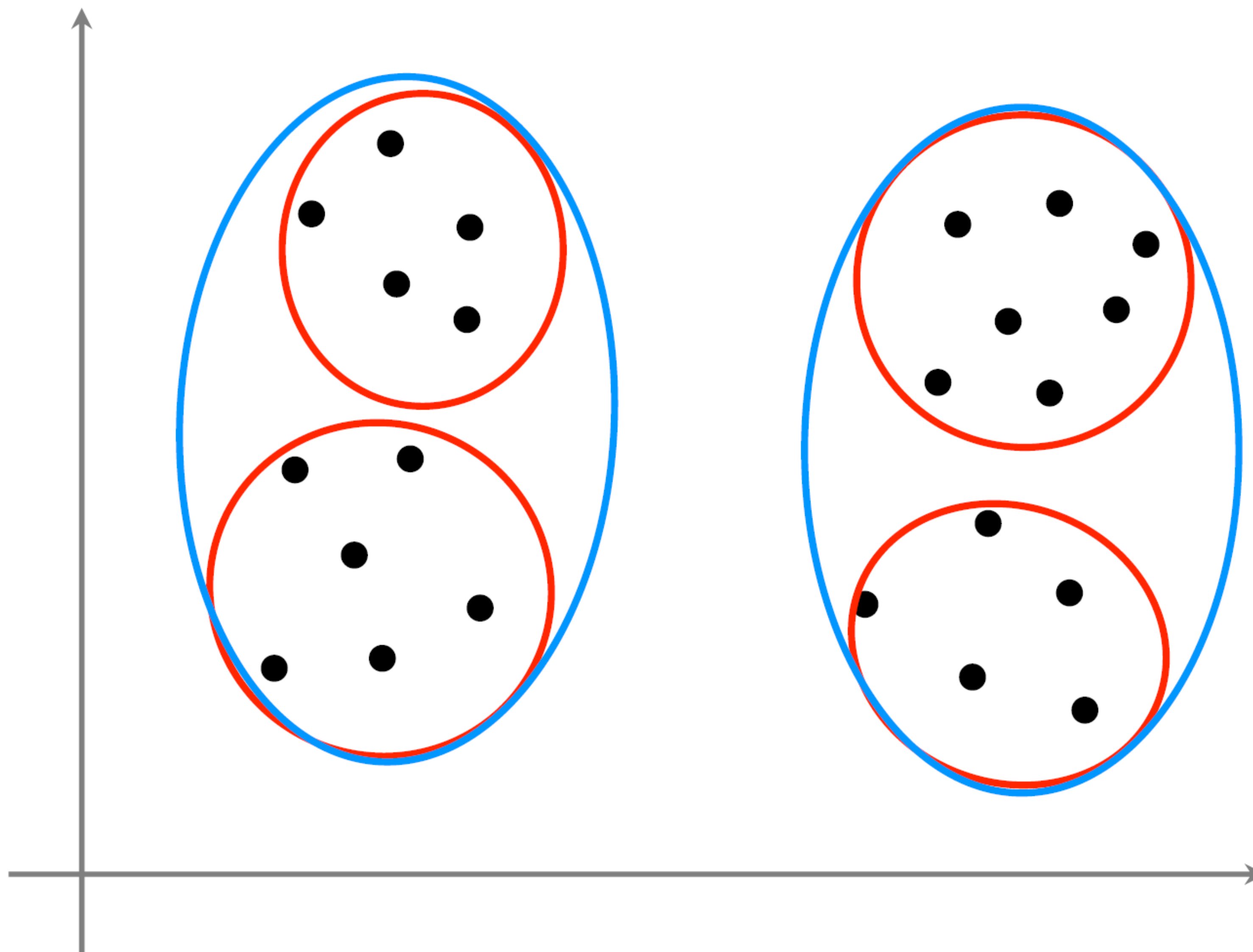
What is the right value of K?



What is the right value of K?



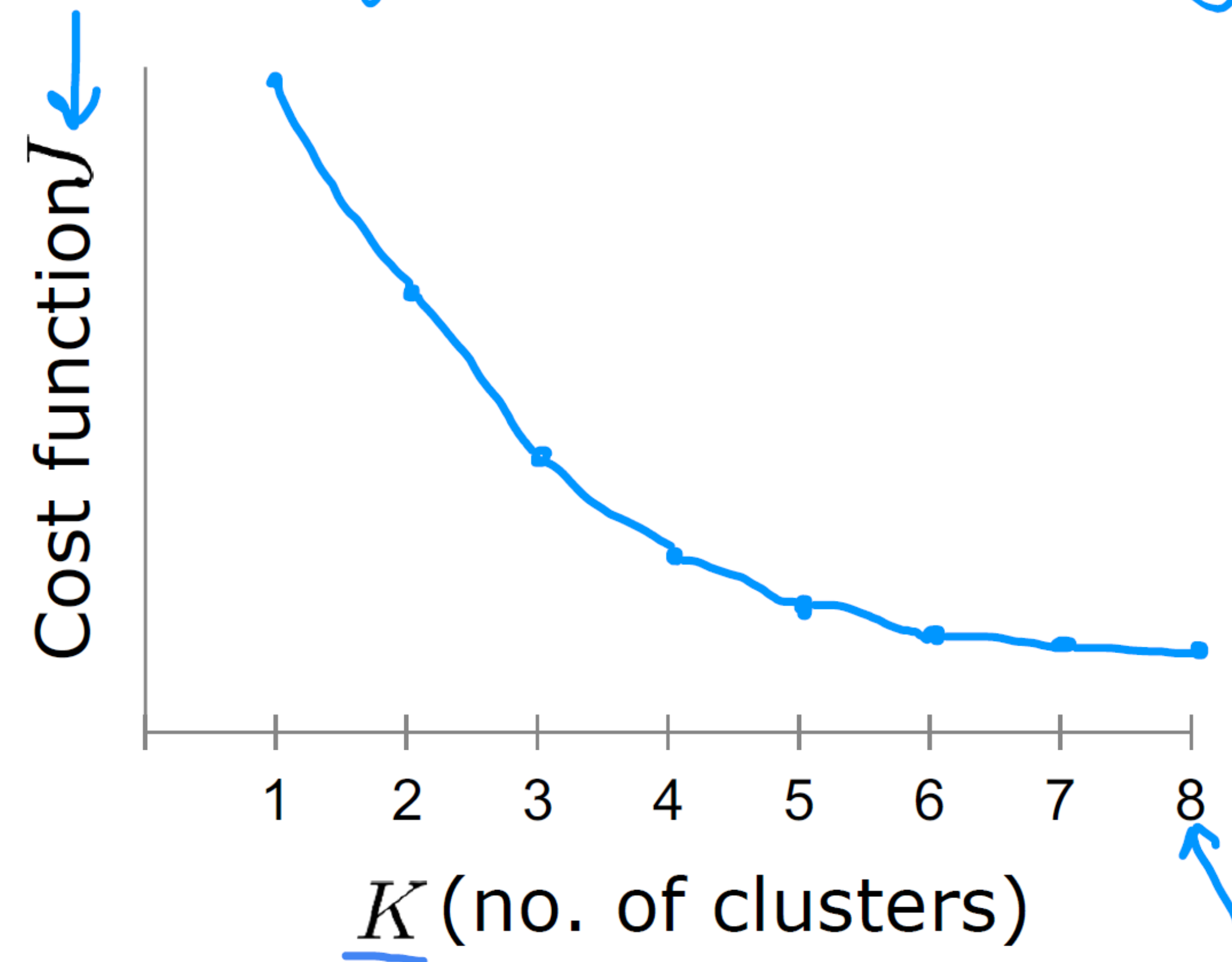
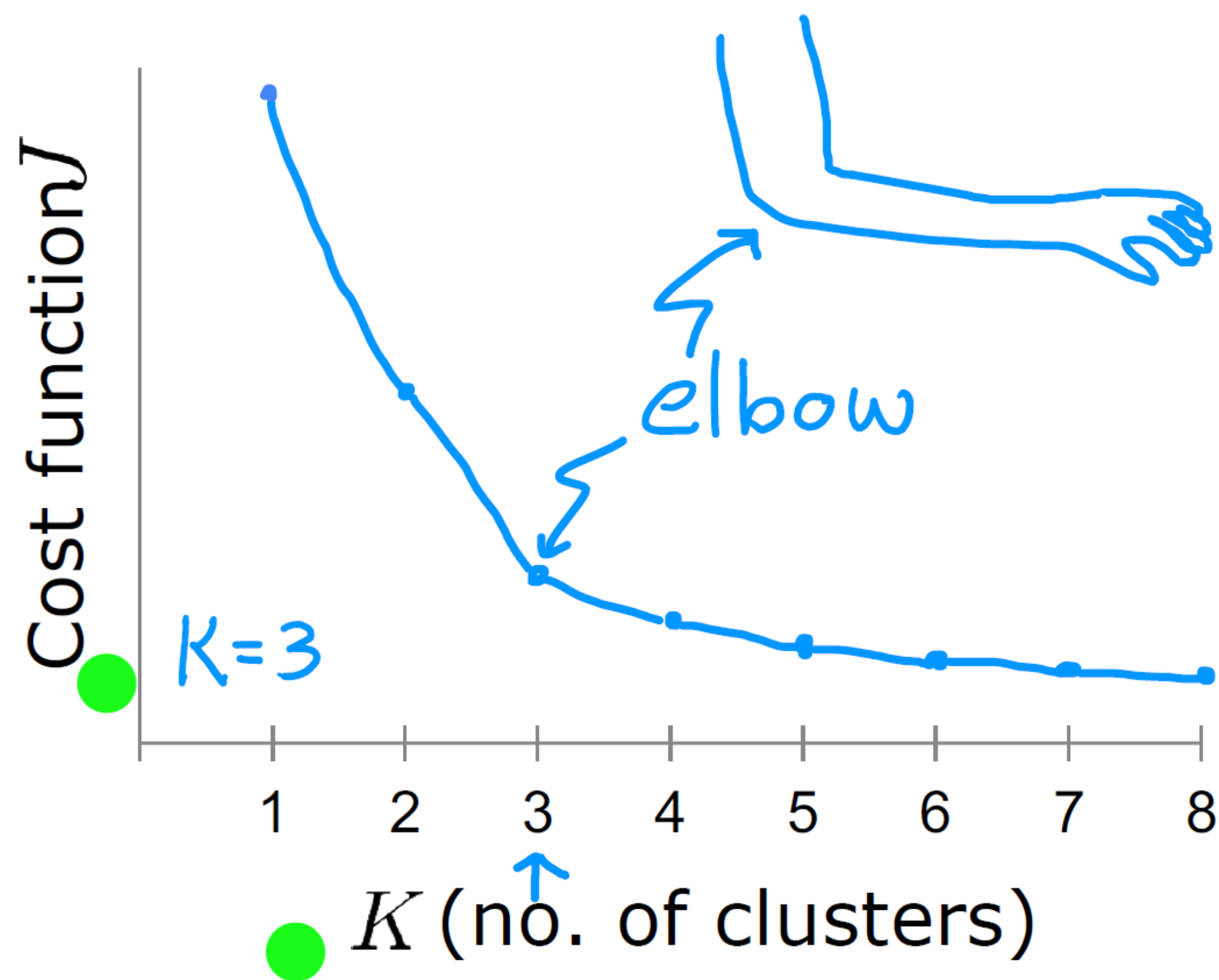
What is the right value of K?



Choosing the value of K

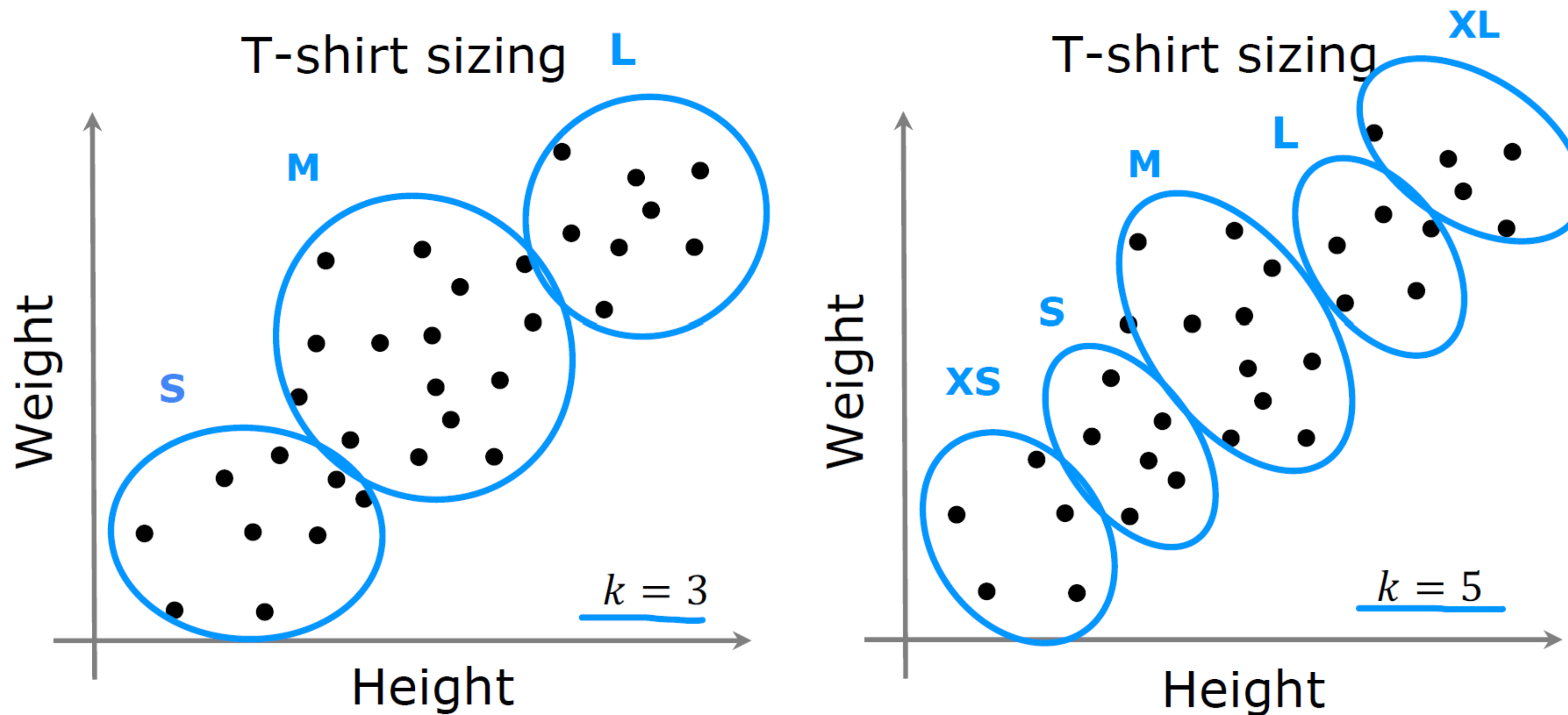
Elbow method

the right " K " is often ambiguous
Don't choose K just to minimize cost J



Choosing the value of K

Often, you want to get clusters for some later (downstream) purpose. Evaluate K-means based on how well it performs on that later purpose.



Limitations

- Need to specify k , the *number* of clusters, in advance (there are ways to automatically determine the best k)
- Sensitive to *noisy data and outliers*
- Not suitable to discover clusters with *non-convex shapes*

Limitations

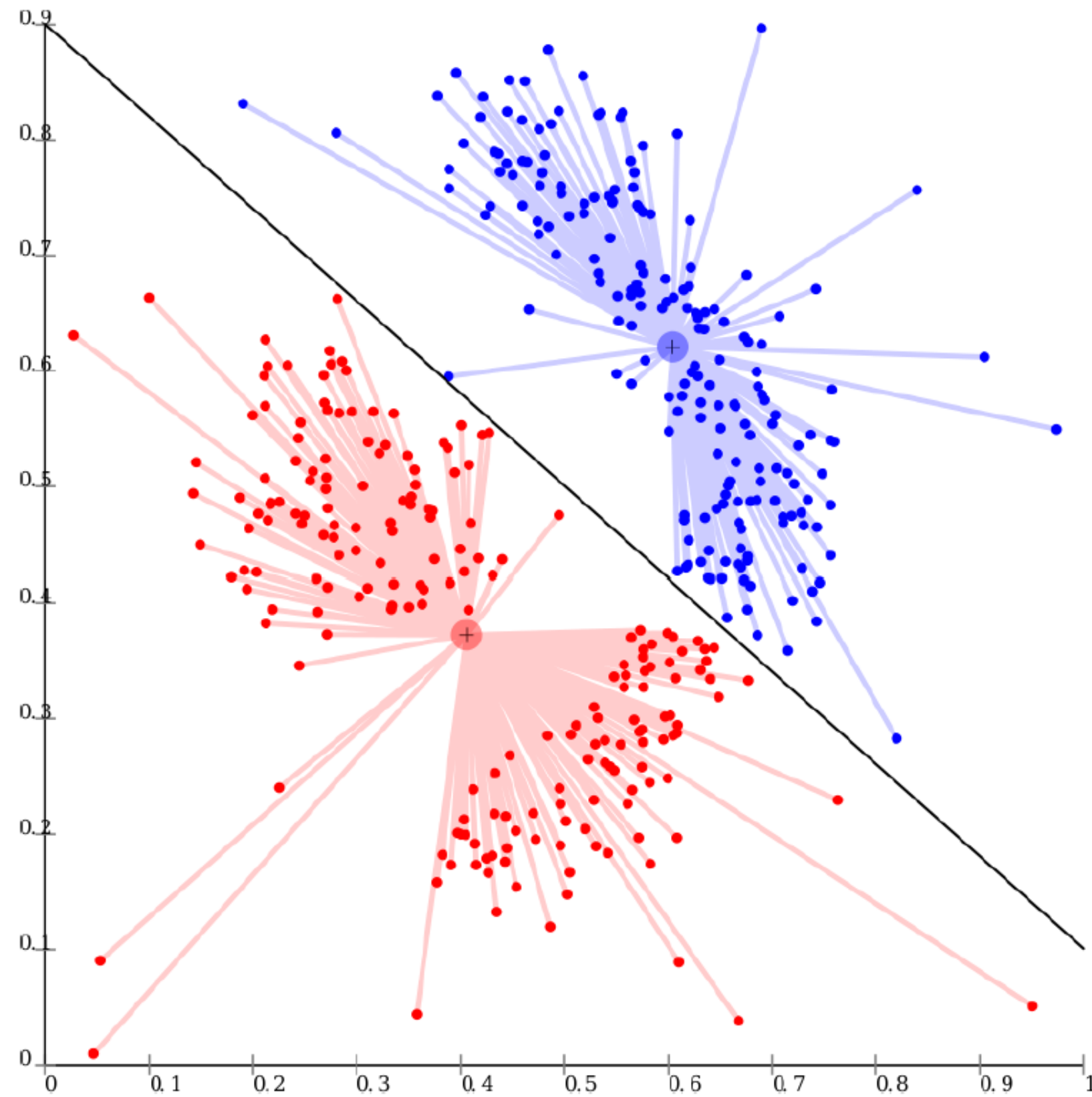


Fig: Convex Shaped

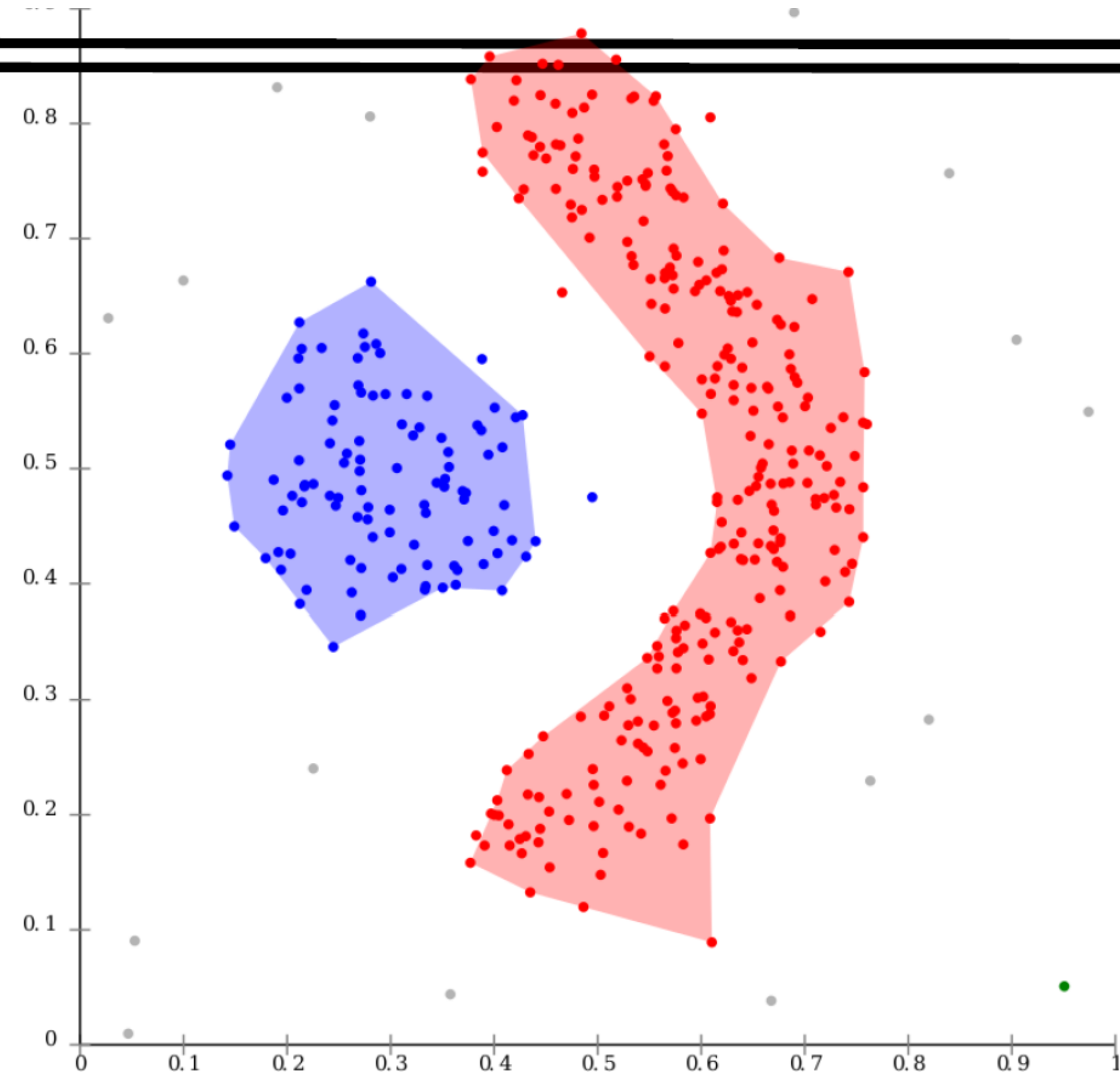


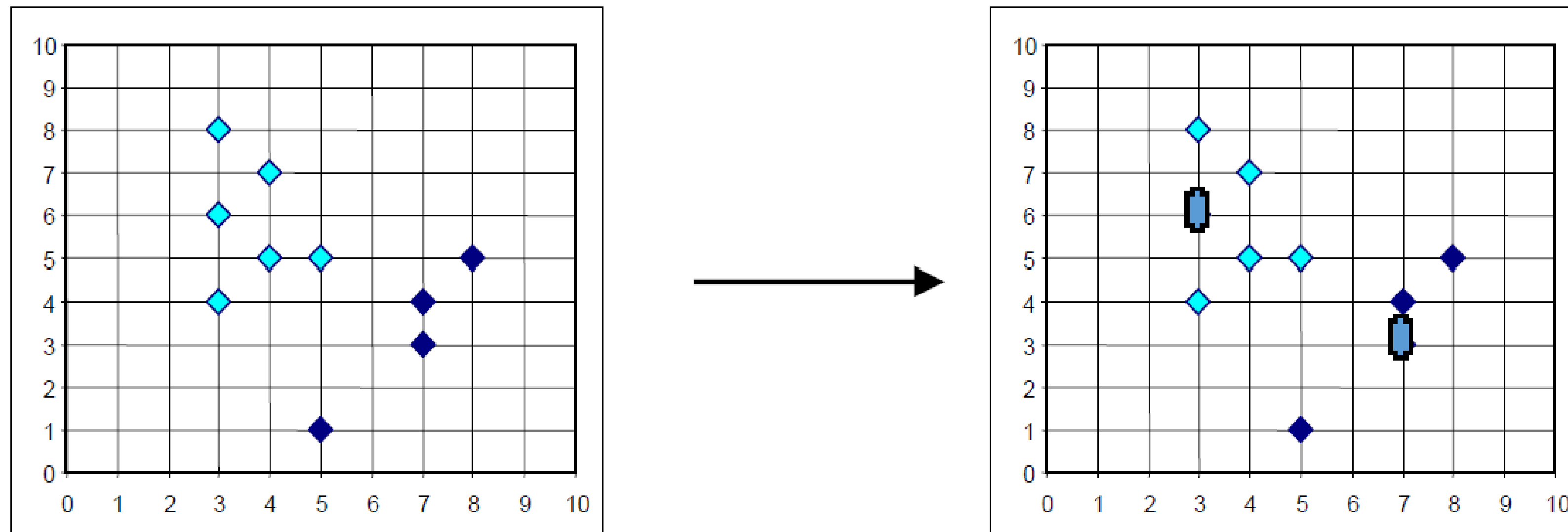
Fig: Non-Convex Shaped

Limitations

The k-means algorithm is sensitive to outliers !

- Since an object with an extremely large value may substantially distort the distribution of the data

K-Medoids: Instead of taking the **mean** value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster



Thank You

Any questions?

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Source & Special Thanks to Andrew Ng (Coursera)
Machine Learning Course 3

<https://www.youtube.com/@Deeplearningai>